

Subjectivity-Intersection Braid Quantum Gravity

A Constructive Refutation of Objectivist Primacy from Source-Derived Quantum Dynamics to the Einstein Limit

Satoru Watanabe

Subjectivity Intersection Emergence Lab (SIEL)

Full working preprint v0.92 — 26 September 2026

Claim status. This is a mathematically explicit *quantum-gravity theory candidate*, not an empirically confirmed theory. In one fixed pointed-Braid model, the source supplies an operational four-direction event cylinder, a Lorentz carrier, a scoped local recorded CTP parent, a four-component history-dressed total Ward identity, and a coefficient-free finite matter-to-geometry-to-matter feedback law. Independently, the declared long-wavelength, local, second-order, formally self-adjoint, conservative class yields $G_{\mu\nu} = (3/5)T_{\mu\nu}$, and a massless spin-2 sector with two physical helicities. Version 0.91 reconstructed finite complex quantum kinematics and an exact low-normalization actuator. Version 0.92 adds a real-QPU implementation of a phase-frame source generator, all-real-time energy-norm strong convergence for a two-polarization hierarchical wave carrier, and a source-derived Higgs/Yukawa/fermion programme with explicit conditional neutrino and Majorana boundaries. The QPU evidence is exploratory, the two polarization slots are not yet identified with the physical ADM/Fierz–Pauli quotient, and a dimensionful Majorana scale is not derived.

Abstract

We develop Subjectivity-Intersection Braid Quantum Gravity as a candidate theory in which a public spacetime carrier is generated by intersections of nonidentical, standpoint-bearing Braid processes rather than assumed as an ontologically prior container. Subjectivity here denotes a located mode of access, transport, or presentation, not consciousness. The construction begins with one pointed quantum-algebraic Braid source and maintains an explicit separation between finite source theorem, continuum limit, physical interpretation, and empirical claim.

The fixed source produces a permutation-covariant rank-four event moment

$$Q_4 = \frac{12}{25}P_0 + \frac{52}{25}P_1$$

and a source-null causal reflection $J_{\text{ray}} = -P_0 + P_1$. Their product $K_{\text{evt}} = J_{\text{ray}}Q_4$ has exact signature $(3+, 1-)$. The same marked event structure selects the full matter corner, ten independent response directions, fixed-model matter routing, and the relative normalization $3/5$ without manual insertion. A refinement clock fixes the Euclidean event-semigroup rate $r = 2\pi/3$ and selects the Lorentzian infrared principal tensor by source continuation rather than a Wick-sign declaration.

The v0.8 advance was a controlled passage from this carrier to a continuous tensor and then to sourced dynamics. Eight typed source sectors carry exact rational $GL(4)$ transport and finite nonmetricity. Braid-derived Gibbs tangents extend them to a smooth event cylinder; the metric, its tangent, and nonmetricity glue over all 24 source charts and transform covariantly under passive local $GL(4)$. Event-path Fisher/Bogoliubov–Kubo–Mori horizontality uniquely lifts metric change to finite positive Markov rates. The lift is flat on all 45 metric two-planes and integrates locally to an exponential leaf. A source-product typing condition then removes the remaining solder ambiguity. Finally, the co-moving causal reflection

$$J(Q, \tau) = I - 2 \frac{(Q\tau)\tau^T}{\tau^T Q \tau}, \quad K(Q, \tau) = Q - 2 \frac{(Q\tau)(Q\tau)^T}{\tau^T Q \tau}$$

restores all ten Lorentz-metric variations without introducing an independent clock coupling. In the declared long-wavelength, local, second-order, formally self-adjoint, conservative class, the finite Braid action variation converges to Hilbert variation, yields an off-shell Noether identity and on-shell Ward conservation, and combines with the source-native first-order gravity sector to give

$$G_{\mu\nu}(g) = \frac{3}{5}T_{\mu\nu}(g, \lambda).$$

Its flat quadratic limit is the massless Fierz–Pauli gauge class with two physical helicities in four dimensions.

Version 0.81 added a finite source-cylinder completion in a narrower, explicitly typed class. The source-fixed raw/Petz record contrast has conditional effect tangents $+D_\alpha/2$ and $-D_\alpha/2$, so its gauge-invariant Choi marginal difference is the complete four-transfer defect $D_\alpha = S_\alpha - \Phi(S_\alpha)$. Placed cellwise on the operational cylinder, this gives a block-local 3 + 1 recorded CTP parent with CPTP normalization, retarded support, positive noise, and exact address/time refinement. The same D_α is the interaction-stress compression in the strong collision intertwiner and yields the exact causal Ward telescope

$$\sum_{k=0}^{n-1} \Phi^k(D_\alpha) = S_\alpha - \Phi^n(S_\alpha)$$

for all four components and all eight source sectors. A full-rank record Fisher action then supplies one finite identity-feedback cycle from matter record to metric register and back to a CPTP matter channel without MMR, CGR, an Einstein target, a manually inserted 3/5, or fitted coefficients.

Version 0.82 reported two further bounded advances. First, the constraint-algebra programme contained exact finite closures and systematic route exclusions through BGCE391. Second, an explicitly derived anomaly-solution transformation groupoid has object algebra $C((\mathbb{Z}_2)^3) \rtimes (\mathbb{Z}_2)^3 \cong M_8(\mathbb{C})$ and Boolean degree multiplicities 1, 3, 3, 1. Its degree-one shell gives three identical copies of the left-Weyl Standard-Model character, with vanishing local anomalies, even Witten parity, and global kernel $\mathbb{Z}_6$. A gauge-equivariant anti-linear real structure leaves one independent complex weak doublet, and the source cubic supplies up-, down-, and charged-lepton Yukawa interaction cycles. This is a scoped construction in a derived extension, not an unchanged-source theorem or an empirical Standard Model derivation.

Version 0.83 reports the first executed calibration layer. A frozen twelve-program, 768-shot AWS Braket Ramsey packet yields an E2-replayed transition frequency $f_{\text{cal}} = 4.484999494698898 \times 10^9$ Hz with standard error $9.314910565190186 \times 10^3$ Hz and $R^2 = 0.9825714110512933$. With the source gap $\Delta h_{\text{cal}} = 4$, this gives device-relative $\tau_* = 1.419442238778493 \times 10^{-10}$ s, $\ell_* = 0.04255380777524273$ m, and $E_* = 7.429480314095178 \times 10^{-25}$ J. Importing the official CODATA 2022 value of G as the second independent calibration input conditionally fixes $\sigma_* = 2.659257381834094 \times 10^{45}$ Pa with partial relative standard uncertainty $2.285490726338230 \times 10^{-5}$. A source audit then proves that public AWS metadata do not provide the provider-clock traceability and systematic uncertainty needed for a complete SI uncertainty budget.

The same version now includes the BGCE444–449 ultraviolet sequence. The rank-ten metric solder induces the exact characteristic-zero decomposition

$$\text{Sym}^2(\text{Sym}^2 V_4^*) \cong \text{Sym}^4 V_4^* \oplus \mathbb{S}_{(2,2)} V_4^*, \quad 55 = 35 + 20,$$

and its source-fixed second response is onto the 35-dimensional quartic sector in all eight source sectors. The finite source-cylinder channels extend uniquely to a strongly continuous UCP semigroup on the AF/quasi-local inductive-limit algebra with complete bound one. The source-native $5^4 = 625$ child rule reconstructs fourth jets exactly on the declared degree-at-most-four polynomial subspace; the actual UB612 PBM metric is connected to the same ordered carrier by a fixed invertible intertwiner. Finally, an exact rational preconditioned interval certificate proves row rank 35 for the actual 35×55 UB612 quartic response throughout its saved coefficient box, with upper bound $0.012750158721617564 < 1$.

Version 0.9 added BGCE450–457. A hierarchical source Laplacian and then an actual-metric positive-conductance deformation generate refinement-compatible UCP heat and positive wave evolution from the same operator, retain anisotropy, and preserve spatial spectral dimension three. Independently, three signed-sector local generators become Braid-specific closable unbounded derivations in the refinement limit; their generated finite-order derivation algebra closes with exact Jacobi identity and no central anomaly in the declared representation. Finally, the operator-logistic identity-feedback law extends the one-cycle construction to every finite iteration: each adaptive step and every finite composition remain CPTP, the nonautonomous history-dressed Ward sum telescopes exactly, and the four source-response directions retain rank four in all eight sectors.

Version 0.91 added BGCE458–491. On the actual 125-element three-history carrier, the oriented cycle $C = R_1 R_2$ obeys $C^3 = I$ and generates a rank-72 real projector P with $J = (C - C^{-1})/\sqrt{3}$ satisfying $J^2 = -I$ on $\text{im } P$, hence a 36-dimensional complex carrier without assuming complex quantum amplitudes. Source reversal, equality counting, and cup/cap gluing then fix normalized squared-norm probabilities in the declared finite source-dagger projective class. Endpoint exchange and the source

Brauer crossing/cap family reduce the observable decomposition to an exact $\mathbb{Z}_2$ superselection rule, and the generated dagger algebra is

$$M_{18}(\mathbb{C}) \oplus M_{18}(\mathbb{C}).$$

The same source supplies four noncommuting compressed controls, a six-order mixed-unitary CPTP engine of Kraus rank six, an exact carrier reflection, coefficient-free LCU block encodings, a source-word select, and a raw- G_1 encoded signal qubit with continuous $SU(2)$ control. Finite spectral interpolation gives exact target pulses for all 32 sector/control cases. The first Lagrange-LCU realization is exact but has success probabilities from approximately $10^{-45.89}$ to $10^{-88.47}$, exposing a black-box efficiency NO-GO. BGCE490 changes basis before amplification: a source-compatible Chebyshev walk reduces the coefficient one-norm to $10^{0.5304} - 10^{3.2497}$ and yields exact deterministic circuits using 7–2,793 phase-matched base calls and 203–39,102 H -block-encoding queries. BGCE491 then hermitianizes the existing source-word LCU unitary and reflects its success address, reproducing the same Chebyshev functional calculus without a square-root-defect or spectral-read primitive.

Version 0.92 adds three bounded advances. First, a 34-CNOT phase-frame representative compiled to 61 native CZ gates and was executed in four 256-shot tasks on Rigetti Cepheus. The frozen sample-level gates passed; a post hoc exact audit retained decisive workspace and two coherence contrasts but did not certify the BASIS modal claim. Second, exact refinement intertwining yields

$$\sup_{t \in \mathbb{R}} \|W(t)x - J_m W_m(t) Q_m x\|_{\mathcal{E}} = \|(I - Q_m)x\|_{\mathcal{E}} \longrightarrow 0$$

for two polarization copies of the mean-zero hierarchical source-cylinder wave carrier. This is stronger than finite-time convergence, but it is not yet physical ADM/Fierz–Pauli convergence. Third, the derived matter route now contains a source-selected three-generation response operator, a nonzero finite Higgs minimum, variational charged Yukawa structure, device-relative bare and dressed mass spectra, and a conditional three-neutral-carrier extension. Normalized unital Frobenius structure fixes $y_v = +1$ only inside the stated completion; exact countermodels show that the existing clock and normalized Majorana pair do not fix a dimensionful Majorana action scale.

The result remains scoped. The finite parent uses the aligned four-score and block-local collision class rather than full ten-component nonlinear metric dynamics. Identity feedback and the metric-conductance map are not proved unique among all admissible dynamics. The direct Fisher coefficient 1 and routing ratio 3/5 are now reconciled as differently typed exact quantities, not equated. The physical ADM projector, smooth Fierz–Pauli principal-symbol identity, nonlinear Euler convergence, provider-clock traceability, observed particle labels, RG running, total uncertainty, and an empirical gravity discriminator remain open. The results refute the universal necessity thesis of *objectivist primacy* by construction; they do not refute mind-independent reality, establish phenomenal consciousness, or show that nature realizes the model.

Keywords: Subjectivity-Intersection Braid Quantum Gravity; quantum gravity candidate; pointed braid; generated objectivity; objectivist primacy; Lorentz signature; emergent spacetime; Einstein equation; metric-affine gravity; Ward identity.

1 What changed from v0.9

Version 0.9 preserved the full finite Lorentz parent, Ward law, Einstein limit, derived Standard-Model representation sector, real-hardware calibration, the BGCE444–456 all-scale package, the BGCE443 scoped constraint algebra, and the BGCE457 all-finite-iteration backreaction theorem. Version 0.91 added the finite quantum reconstruction and exact low-normalization control layer through BGCE491. Version 0.92 retains that full manuscript and adds the post-v0.91 physical compiler, real-QPU, continuum-dynamics, and matter-spectrum results through BGCE569 and BQGEULER-004. It does not silently promote exploratory QPU evidence, hierarchical convergence, or a conditional neutrino completion into empirical or unconditional quantum-gravity claims.

The strongest defensible claim is upgraded but remains bounded. The former finite-parent and four-current gaps, the all-refinement channel and actual quartic-response subproblem, the source-generated local-derivation constraint algebra, anisotropic hierarchical UCP carrier, all-finite-iteration adaptive back-reaction, and finite source-dagger quantum reconstruction are each closed in their recorded scopes. The

phase-frame source generator has now crossed from abstract compilation to exploratory real-QPU execution, while BQGEULER-004 closes all-real-time strong convergence only on the hierarchical two-slot carrier. The unrestricted SDPC claim is not declared solved because full ten-component nonlinear dynamics, physical ADM/Fierz–Pauli identification, a source-forced unique feedback or conductance law, fully traceable calibration, observed matter identification, a competitor-divergent gravity observable, and empirical confirmation remain incomplete.

2 Subjectivity intersection and the target of the refutation

2.1 Subjectivity is not consciousness

A *subjectivity* in this work is a typed and located mode of access, transport, or presentation whose distinction from other such modes is constitutive of the construction. It may be represented by a source prefix, a reference system, or an oriented transport context. No braid, photon, clock, algebra, or source sector is thereby attributed sensation, awareness, intention, or phenomenal experience. Any bridge from this formal notion to consciousness would require a separate axiom and separate evidence.

Let A and B denote nonidentical source standpoints. Their intersection does not mean a third external observer that views both from nowhere. It is the compositional event in which their difference is retained while presentation redundancy is removed. The public structure $C(A, B)$ is then an output of the intersection rather than an independently completed container.

Definition 2.1 (Generated objectivity). A carrier $C(A, B)$ exhibits generated objectivity when it is invariant under allowed internal presentation changes, covariant under ambient re-description, symmetric under exchange of the two typed standpoint roles when the source law requires it, altered by the deletion of either role, and regenerated by one source law through the declared refinement stages without stage-dependent fitting.

2.2 The exact objectivist claim being denied

Definition 2.2 (Objectivist Primacy Thesis, OP). Every mathematically admissible construction of a stable, shareable, transformation-respecting physical carrier must take that carrier, or an equivalent perspective-neutral carrier, as primitive. Distinct standpoints may describe it, but cannot generate it.

OP is a universal necessity thesis. It is stronger than scientific reproducibility, realism about measured regularities, covariance, or mind-independent existence. A single valid generated-objectivity model refutes OP as stated. It does not prove that the model is realized in nature. The present paper therefore keeps two conclusions separate:

1. a mathematical counterexample to the necessity of objectivist primacy; and
2. a physical quantum-gravity hypothesis whose empirical truth remains open.

The proposal is adjacent to, but not identical with, relational quantum mechanics, quantum-reference-frame programmes, QBism, perspectival realism, and structural realism [11, 5, 14, 3, 10, 8, 15]. Its distinct technical claim is generative: a shared tensor carrier is reconstructed from maintained source provenance and intersection rather than presupposed as a perspective-neutral starting object.

3 Pointed-Braid source and operational event cylinder

3.1 Quantum-algebraic source

At refinement stage n , take the matrix algebra

$$\mathcal{A}_n = M_{5^n}(\mathbb{C}), \quad j_n(X) = X \otimes I_5. \quad (1)$$

The source contains distinguished states, marked local algebras, and unitary Braid transports satisfying the typed Yang–Baxter and composition laws. This is quantum in the algebraic sense: states, complex noncommutative observables, tensor composition, and unitary transport are present. The present paper does not derive complex amplitudes, the Born rule, or Hilbert-space composition from an unstructured topological braid group.

The source data are therefore neither “bare braid topology” [1] nor a continuum spacetime. They are a finite quantum-algebraic pointed-Braid source developed in the preceding Subjectivity-Intersection Mathematics programme [16, 17]. The scientific question is how much geometry and dynamics this source forces without target fitting.

3.2 Operational event identification

The fixed source packet contains a marked event effect P and a spectral event algebra. The actual stored event transition R_{evt} acts on 25 local atoms, is involutive, and satisfies the declared Yang–Baxter event relation. The same marked generator determines four future-null affine links. This identification retires an independent formal correspondence principle CGR inside the fixed model: the event carrier is read from the source law itself rather than attached as a separate map.

This is a scoped statement. It does not say that every operational event in nature must be represented by this P , nor does it select a natural signed sector. It says that the previously separate formal event correspondence is redundant in the declared source construction.

3.3 Four dimensions without a primitive completed spacetime

The operational cylinder has four source directions because four independently certified null links span the event carrier. Let L be the matrix with these links as columns. The exact certificate gives

$$\det L = \frac{81}{16} \neq 0. \quad (2)$$

Thus the event carrier is four-dimensional in the fixed model. This is stronger than postulating an arbitrary ambient $\mathbb{R}^4$ and then placing Braid data inside it: the four operational directions are generated and are nondegenerate. It is weaker than a theorem that four dimensions are inevitable for all admissible Braid sources. The latter is not claimed.

4 Matter routing and the non-manual factor 3/5

4.1 The full event-conditioned corner

The marked event defines the physical corner candidate

$$\mathcal{A}_\cap = PAP, \quad (3)$$

with Hamiltonian $H_\cap = PG_1P$ and ten compressed source operators

$$F_A^\cap = PF_AP, \quad A = 1, \dots, 10. \quad (4)$$

The corresponding Gibbs family is faithful on the corner, and its Bogoliubov–Kubo–Mori Hessian has rank ten. Hence the ten declared matter-response directions are locally nondegenerate.

The source atoms generate five partition retractions. Requiring exact preservation of the already fixed eleven-operator packet

$$\mathcal{S}_\cap = \{PG_1P, PF_1P, \dots, PF_{10}P\} \quad (5)$$

selects only the full partition. Every nontrivial dephasing changes $H_\cap$ before any coefficient can be retuned. This closes MMR2 within the declared fixed event-conditioned model and complete five-partition class. It does not prove uniqueness among all completely positive instruments or all possible subalgebras.

4.2 Where 3/5 comes from

The normalized tail trace sends the corner unit to the outer event effect P . Its source-cup composition yields the relative multiplier

$$\kappa_B = \frac{3}{5}. \quad (6)$$

No line of the Lorentz-carrier or backreaction calculation sets or retunes this value. It is inherited from the same marked source routing that selects the event-conditioned matter packet. In particular, the appearance of the same coefficient in the Einstein equation is not counted as proof of that equation: source-derived normalization, tensor typing, conservation, and common metric variation are proved as separate gates.

5 Braid-derived local-label dynamics

5.1 Actual event transition versus identity and flip

For each adjacent pair of source directions, define the displacement under the actual Braid event transition. Its exact pair moment is

$$M_{\text{pair}} = \begin{pmatrix} 28 & -20 \\ -20 & 28 \end{pmatrix}, \quad (7)$$

with eigenvalues 8 and 48 and rank two. The identity comparator has rank zero and the ordinary flip comparator rank one. Thus the rank-two coupling is specific to the actual non-flip event map within this comparator class.

Average the adjacent-pair moment over the 24 source charts. The result is

$$Q_4 = \frac{104I_4 - 20\mathbf{1}\mathbf{1}^\top}{50} = \frac{12}{25}R_0 + \frac{52}{25}R_1, \quad (8)$$

where

$$R_0 = \frac{1}{4}\mathbf{1}\mathbf{1}^\top, \quad R_1 = I_4 - R_0. \quad (9)$$

Q_4 has diagonal entries $42/25$, off-diagonal entries $-2/5$, and eigenvalues $12/25$ on the uniform line and $52/25$ on its three-dimensional complement. It is positive and full rank. No distinguished physical $1 + 3$ ordering or external axis weights enter Eq. (8).

5.2 Relative entropy and the BKM–Dirichlet form

The source response is the derivative of a log-partition function. Within the response-exact Bregman class this selects the log-Z Bregman divergence, equivalently the Umegaki relative entropy for the Gibbs family [13, 2]. On actual neighbor event edges, the quadratic term is

$$\delta^2 S_{\text{edge}} = \frac{1}{2} \int Q_4^{ab}, \langle \partial_a \lambda, \partial_b \lambda \rangle_{\text{BKM}}, d\mu. \quad (10)$$

The base moment has rank four and the internal BKM metric rank four in the tested local-label sector, so the gradient quadratic form has rank sixteen. This supplies source-independent local-label dynamics on the fixed cylinder. It is a positive Dirichlet form, not yet a Lorentzian action.

6 Null tetrad and source-derived Lorentz carrier

6.1 The source-null metric

The four null links have a source-fixed Gram matrix G_{null} with zero diagonal and off-diagonal entries $-12/25$. Since L is invertible, the ambient metric reconstructed from the links is unique:

$$g = L^{-\top} G_{\text{null}} L^{-1} = \text{diag}\left(-\frac{16}{25}, \frac{16}{75}, \frac{16}{75}, \frac{16}{75}\right). \quad (11)$$

It has signature $(3+, 1-)$. The induced map on symmetric tensors, $\text{Sym}^2(L)$, has rank ten. Thus the four-direction carrier is algebraically capable of supporting all ten symmetric rank-two components.

This statement alone does not solve the physical stress problem. Full rank is an existence fact about an algebraic map; it does not prove that the actual matter operator packet transforms through that map or that the resulting response is conserved.

6.2 Why a coframe alone cannot create the Lorentz sign

The positive event moment remains positive under the null coframe:

$$LQ_4L^\top = \text{diag}\left(\frac{27}{25}, \frac{117}{25}, \frac{117}{25}, \frac{117}{25}\right). \quad (12)$$

By Sylvester’s law of inertia, no real invertible congruence changes signature $(4+, 0-)$ to $(3+, 1-)$. Therefore a claim that “the null tetrad automatically Lorentzianizes the positive edge action” is false. This no-go forces the causal sign to have a source origin other than coordinate change.

6.3 Causal grading

The source causal analysis distinguishes the uniform sum of the four future-null rays as the timelike direction. The corresponding reflection is

$$J_{\text{ray}} = I_4 - \frac{1}{2} \mathbf{1}\mathbf{1}^\top = -P_0 + P_1. \quad (13)$$

It has eigenvalue -1 on the uniform line and $+1$ on the standard three-dimensional complement. Since Q_4 is S_4 -equivariant, J_{ray} and Q_4 commute. Hence

$$K_{\text{evt}} = J_{\text{ray}} Q_4 = -\frac{12}{25} P_0 + \frac{52}{25} P_1 \quad (14)$$

is symmetric with exact signature $(3+, 1-)$. In components its diagonal is $36/25$ and every off-diagonal entry is $-16/25$.

Theorem 6.1 (Source-derived Lorentz event carrier). *In the declared fixed source model, the actual Braid event moment Q_4 and the independently source-selected causal grading J_{ray} determine the Lorentz-signature carrier $K_{\text{evt}} = J_{\text{ray}} Q_4$. The construction uses no Wick rotation, manual sign assignment, or fitted continuous coefficient.*

6.4 Natural normalization to the null Gram tensor

Because Eq. (14) is contravariant in the kinetic form, its physical comparison target is G_{null}^{-1} , not G_{null} . Restrict a normalizer to the positive, self-adjoint, S_4 -equivariant class

$$A_{\text{kin}} = a_0 P_0 + a_1 P_1. \quad (15)$$

Then

$$A_{\text{kin}}^T K_{\text{evt}} A_{\text{kin}} = G_{\text{null}}^{-1} \quad (16)$$

fixes

$$a_0 = \frac{25\sqrt{3}}{36}, \quad a_1 = \frac{25\sqrt{39}}{156} \quad (17)$$

uniquely within that class. All 24 chart permutations commute with the projector decomposition. In the character frame one recovers

$$g^{-1} = \text{diag}\left(-\frac{25}{16}, \frac{75}{16}, \frac{75}{16}, \frac{75}{16}\right). \quad (18)$$

The uniqueness claim is deliberately local: it is not uniqueness under arbitrary Lorentz isometries or among all nonlinear field redefinitions.

7 From a Lorentz carrier to a Lorentz action

7.1 Why the action question is separate

The positive Umegaki edge functional underlying Eq. (10) has Hessian $Q_4 \otimes G_{\text{BKM}}$. The causal carrier has principal tensor $K_{\text{evt}} \otimes G_{\text{BKM}}$. Since the latter contains one negative mode, equality of the two cannot be obtained by a real invertible coframe congruence. A carrier identity is therefore not yet an action theorem.

Three tempting shortcuts fail exactly. First, the canonical Hamiltonian-orbit action generated by G_1 is first order in velocity and has zero velocity Hessian. Its existing metric response has rank three, not four. Second, forward and reverse event displacements have the same second moment; signed orientation averages vanish and scalar combinations remain scalar multiples of Q_4 . Third, the doubled closed-time-path form

$$\eta_{\text{br}} \otimes Q_4, \quad \eta_{\text{br}} = \text{diag}(1, -1), \quad (19)$$

has signature $(4+, 4-)$ on branch $\otimes$ event space, while any fixed branch contraction is again only a scalar multiple of Q_4 . None selects the one-dimensional clock line inside the four-dimensional event carrier.

A graded trace does perform the algebraic conversion,

$$\text{Str}_J(X) = \text{Tr}(J_{\text{ray}} X), \quad \text{Str}_J(Q_4) = J_{\text{ray}} Q_4, \quad (20)$$

and is unique up to scale among graded-cyclic traces fixed by this $\mathbb{Z}_2$ grading. It is not positive because $\text{Str}_J(P_0) = -1$. Declaring a change from ordinary positive operator algebra to a graded category would therefore make a conditional construction, not a Braid-only derivation.

7.2 Positive causal sectors and source continuation

The same Lorentz Hessian can be written as a difference of ordinary positive sector Hessians,

$$H_0 = \frac{12}{25}P_0 \otimes G_{\text{BKM}}, \quad H_1 = \frac{52}{25}P_1 \otimes G_{\text{BKM}}, \quad H_1 - H_0 = K_{\text{evt}} \otimes G_{\text{BKM}}. \quad (21)$$

Its signature is $(12+, 4-)$ when the internal BKM tangent has rank four. This removes the need for a nonpositive trace but still leaves a continuation rule between the positive source functional and the causal difference.

The source causal reflection satisfies

$$J_{\text{ray}}^\top Q_4 J_{\text{ray}} = Q_4, \quad J_{\text{ray}}^\top Q_4^{-1} J_{\text{ray}} = Q_4^{-1}. \quad (22)$$

The raw 25-event involution P consists of nine fixed points and eight transpositions. Hence P has eight eigenvalues -1 and cannot be represented as e^{-aH} with $H \geq 0$. The positive generator is instead the graph Laplacian

$$L_{\text{evt}} = I - P, \quad K_t = e^{-rtL_{\text{evt}}}. \quad (23)$$

The marked source refinement interval $\Delta\tau = 3/4$ and the minimal Stone endpoint π fix

$$r = \frac{2\pi}{3}. \quad (24)$$

There is therefore no fitted event rate. Reflection continuation of the source clock line changes the uniform sector sign and leaves the three-dimensional complement positive.

Theorem 7.1 (Scoped source selection of the infrared Lorentz action). *Within the declared real Stone/affine completion of the fixed marked source, the event semigroup with rate (24), the source reflection, and the response-exact BKM edge functional select the infrared quadratic tensor*

$$K_{\text{evt}} \otimes G_{\text{BKM}}. \quad (25)$$

No manual Wick sign, external rate, or continuous coefficient fit is required.

The theorem is an infrared principal-symbol theorem. It does not make the unmodified full finite Umegaki functional Lorentzian, a distinction that becomes decisive in Section 13.

8 Quantum channel and finite metric-affine bundle

8.1 A source-native quantum channel on the same carrier

For each of the eight source masks, let R_m denote the normalized ordered-edge Gram matrix. Each R_m is exactly positive definite with unit diagonal. The Schur map

$$\Phi_m(X) = R_m \circ X \quad (26)$$

is consequently completely positive, trace preserving, and unital. It is nontrivial and admits a Stinespring representation. The source-null tetrad normalized by its source norm gives

$$U = \frac{2}{3}L = \frac{1}{2}H_4, \quad (27)$$

an orthogonal Hadamard transformation. It sends the character clock projector to the ray clock projector and intertwines the source channel with

$$P_0, \quad J_{\text{ray}}, \quad Q_4, \quad K_{\text{evt}}. \quad (28)$$

Thus a standard physical quantum operation and the Lorentz causal carrier live on one typed four-carrier rather than on separately postulated state spaces.

This result does not identify a derivative. The eight endpoint Grams are distinct while the anchor K_{evt} is common. Linear and quadratic positive paths joining the same endpoints have different initial derivatives. Endpoints alone cannot select a tangent, Hilbert stress, or Ward law.

8.2 Direct rational carrier transport

Let C_m be the unique four-column carrier matrix in source sector m . The source-selected least-squares transport is

$$T_{m \leftarrow n} = (C_m^\top C_m)^{-1} C_m^\top C_n. \quad (29)$$

Every one of the 48 oriented source-cube packets is rank four and therefore invertible. The ordered-edge Gram is not preserved:

$$N_{mn} = T_{m \leftarrow n}^\top G_m T_{m \leftarrow n} - G_n. \quad (30)$$

All 48 defects are symmetric and rank four at both fixed good primes. This is a finite metric-affine structure: connection and nonmetricity are outputs of the same source.

Equation (30) must not be divided directly by the spacetime mesh. Its endpoints are internal source sectors, not neighboring spacetime cells. The type error was explicitly tested and rejected. Continuous differentiation instead comes from the already existing Gibbs source coordinates.

8.3 Characteristic-zero tangent theorem

Let θ_i , $i = 1, 2, 3$, denote the source-local Gibbs labels. Differentiating the carrier module and the ordered-edge Gram along these labels yields

$$\dot{C}_{m,i} = 0, \quad A_{m,i} = 0, \quad N_{m,i} = \dot{G}_{m,i}. \quad (31)$$

The equalities for the carrier and its projection connection were checked as exact rational identities: 288 of 288 carrier commutators vanish across eight typed-gauge-inequivalent sectors and three label directions. In contrast, every one of the 24 Gram tangents is nonzero, symmetric, and rank two. Hence

$$N_{m,i} = \dot{G}_{m,i} \neq 0, \quad \text{rank } N_{m,i} = 2. \quad (32)$$

The metric changes while the preferred source gauge keeps the carrier stationary to first order. In a nonconstant frame the connection is, of course, not zero; the inhomogeneous frame term restores the usual tensor law.

9 Continuous event-cylinder tensor field

9.1 Chain rule without an external ambient manifold

The smooth event cylinder was defined before the metric-continuation calculation. Its local label fields are not added for this theorem. Across all eight sectors the three metric responses have rank three. Combining them with the four-dimensional event base gives the rank-twelve chain-rule symbol

$$\partial_a G_m(x) = \sum_{i=1}^3 \frac{\partial G_m}{\partial \theta_i} \partial_a \theta_i(x), \quad \text{rank } \sigma(\partial G) = 4 \times 3 = 12. \quad (33)$$

Equation (33) is the first-order passage from finite source labels to a continuous local field. It does not identify the cylinder with measured natural spacetime; it establishes a source-derived operational base on which differentiation is defined.

9.2 Chart gluing and local $GL(4)$ naturality

The 24 source charts act by the four-ray permutation representation. Metric, metric tangent, and non-metricity obey the same congruence law in every chart. Exact checks include 192 metric packets, 576 tangent/nonmetricity packets, and 13,824 double-overlap compositions. Direct $A \rightarrow C$ transport equals $A \rightarrow B \rightarrow C$ transport.

The finite result extends to arbitrary passive local frame changes. With the source-fixed solder U , a frame Jacobian $J(x)$ acts on the carrier by

$$C_J(x) = UJ(x)U^\top. \quad (34)$$

Writing A for the source-gauge connection, the transformed objects are

$$G' = C_J^\top G C_J, \quad (35)$$

$$A' = C_J^{-1} A C_J + C_J^{-1} dC_J, \quad (36)$$

$$N' = dG' - A'^\top G' - G' A' = C_J^\top N C_J. \quad (37)$$

The representation product, its S_4 restriction, the actual metric, the variable-frame connection, and non-metricity all pass exact composition tests. Thus the result is not merely 24 coincident matrices: it is a local tensor field on the declared event domain.

Theorem 9.1 (Source-event tensor gluing). *The Gibbs-derived ordered-edge metric and its nonmetricity define an S_4 -glued, passively local- $GL(4)$ -natural first-order tensor field on the smooth source event cylinder. No external $\mathbb{R}^4$ coordinate field or fitted gluing coefficient is required.*

Passive covariance alone does not select a matter action. Many curvature-improved actions are covariant and conserved while differing in their Hilbert stress. The next step therefore asks whether the source Markov law fixes active metric dependence.

10 Active Braid transport and the ten metric directions

10.1 Minimal conservative operator

The source cell measure multiplied by the metric volume density transforms naturally under local $GL(4)$. For a local second-order scalar operator with the derived principal symbol and density, formal self-adjointness fixes the four drift coefficients. Markov conservativity fixes the zero-order potential to zero. In particular, an independently added αR improvement is excluded within this declared class because it violates preservation of the constant function.

This is a uniqueness statement within a class, not a theorem that all physical actions must be second order or Markovian. The critical remaining issue is whether a change in the derived metric uniquely changes the actual Braid transition rates.

10.2 Horizontal conductance lift

There are eight nontrivial undirected Braid-event edges and twelve source-typed conductance modes. Their moment map to symmetric metric components has rank ten and a two-dimensional kernel. Every one of the 24 actual metric tangents lies in its image, but the metric alone cannot see the two residual edge coordinates.

The event-path probability law supplies the missing criterion. Equip rate variations with the Fisher/BKM pairing at the source rate $2\pi/3$. Among all rate variations producing a given metric variation, select the vector orthogonal to the kernel. The raw 48-variable kernel has dimension 38 and full BKM rank 38; the residual two-dimensional typed kernel also has BKM rank two. The horizontal lift is therefore unique. It

reproduces 24 of 24 source tangents, is S_4 -covariant, and fixes the common-to-relative quotient weight ratio to 18 : 1 without fitting.

Let M denote the rank-ten moment matrix and r_0 the source rate vector. The horizontal leaf has the finite form

$$r(\xi) = r_0 \circ \exp(M^\top \xi), \quad (38)$$

where the exponential and product are componentwise. Its Jacobian is

$$DM(\xi) = M \operatorname{diag}(r(\xi))M^\top > 0. \quad (39)$$

Hence the metric moment is locally invertible, rates remain positive, and row-sum zero, detailed balance, and refinement compatibility are preserved. The horizontal connection has exact zero curvature on all 45 independent metric two-planes. Equation (38) is consequently a local finite integration, not merely a first-order fit.

10.3 The causal 7 + 3 split

The positive X21R1 Gram is not itself the physical Lorentz metric: neither it nor its inverse matches the positive event moment in any of the eight sectors. The event sector at fixed causal reflection supplies seven J -even metric directions. The three Gibbs-label directions supply precisely the missing J -odd time–space directions. Their combined response has rank ten in all sectors and glues as a tensor.

At this point a subtle variational obstruction appears. If the odd metric variables are identified with the matter labels θ_i , holding matter fixed during a Hilbert variation locks those three directions and leaves rank seven. The remedy cannot be to vary matter while claiming a fixed-matter stress. The source already contains ten matter-independent source variables a_A , but covariance alone is insufficient to solder them: the S_4 commutant on Sym^2 has dimension nine, and clock, grading, and event spectrum still leave a five-dimensional family.

The actual source typing removes this ambiguity. The ten variables are not an arbitrary ten-dimensional representation; they are generated as symmetric products of four carrier variables. Requiring preservation of that product generation and of the already derived four-dimensional map U forces

$$A \mapsto U^\top A U \quad (40)$$

with no residual continuous freedom. This is the physical solder used below.

Proposition 10.1 (Typed solder uniqueness). *Within the existing source-product class, Eq. (40) is the unique rank-ten solder compatible with the four-carrier map, source typing, and chart action. Covariance without the product-typing condition is not sufficient.*

11 Co-moving causal grading, Hilbert stress, and Ward identity

11.1 The fixed-grading obstruction

A fixed J_{ray} transports only seven of the ten positive-form variations into symmetric Lorentz-metric variations. This caused an explicit reversal of an earlier stress/Ward completion claim. The missing three directions are the time–space mixtures. They are recovered only if the clock projector moves with the positive event form.

Let τ be the source clock ray, normalized only projectively. Define

$$J(Q, \tau) = I - 2 \frac{(Q\tau)\tau^\top}{\tau^\top Q \tau}, \quad (41)$$

$$K(Q, \tau) = J(Q, \tau)Q = Q - 2 \frac{(Q\tau)(Q\tau)^\top}{\tau^\top Q \tau}. \quad (42)$$

At the source anchor these recover J_{ray} and K_{evt} exactly. The off-shell Jacobian $D_Q K$ has rank ten. The clock derivative has rank three but is not an independent coupling:

$$D_\tau K = (D_Q K) C, \quad (43)$$

for an exact source-derived coefficient map C , while the clock-scale direction lies in the kernel. Moreover, the clock reflection is self-inverse on the tested positive-form domain,

$$K = \mathcal{R}_\tau(Q), \quad Q = \mathcal{R}_\tau(K). \quad (44)$$

Thus the clock is a redundant local parametrization of the full-rank Lorentz tensor, not a new continuum field inserted to restore rank.

11.2 Continuum matter action and stress

Let λ^A denote the matter coordinates and $G_{AB}(\lambda)$ the BKM metric inherited from the full source corner. The source-selected long-wavelength action in the declared class is

$$S_m[g, \lambda] = \frac{1}{2} \int_{\mathcal{M}} d^4x \sqrt{-g} g^{\mu\nu} G_{AB}(\lambda) \partial_\mu \lambda^A \partial_\nu \lambda^B. \quad (45)$$

This is not an independently guessed nonlinear sigma model. Its local density, principal tensor, internal BKM pairing, edge coupling, active rate dependence, and metric solder are the continuum images of the finite source construction described above.

Variation through the independent source variables a_A and the unique solder gives the Hilbert tensor

$$T_{\mu\nu} = G_{AB} \left(\partial_\mu \lambda^A \partial_\nu \lambda^B - \frac{1}{2} g_{\mu\nu} g^{\rho\sigma} \partial_\rho \lambda^A \partial_\sigma \lambda^B \right). \quad (46)$$

Diagonal diffeomorphism covariance yields the off-shell identity

$$\nabla^\mu T_{\mu\nu} = -\mathcal{E}_A(\lambda) \partial_\nu \lambda^A, \quad (47)$$

where $\mathcal{E}_A = 0$ are the matter Euler–Lagrange equations. Therefore

$$\mathcal{E}_A = 0 \implies \nabla^\mu T_{\mu\nu} = 0. \quad (48)$$

Theorem 11.1 (Scoped Braid-source stress/Ward theorem). *For the fixed source, and within the long-wavelength, local, second-order, formally self-adjoint, Markov-conservative class, the finite response-exact Braid action, horizontal active transport, source-product solder, and co-moving causal reflection determine Eq. (45). Its fixed-matter variation gives Eq. (46), and its source symmetry gives the off-shell and on-shell Ward laws (47).*

The restrictions in Theorem 11.1 are noncompensating. The theorem does not cover arbitrary curvature improvements, higher derivatives, nonlocal actions, or the full finite dissipative quantum evolution. Within its declared class, however, no separate MMR, stress solder, Ward postulate, causal-sign insertion, or coefficient fit remains.

12 Noncircular backreaction and the Einstein limit

12.1 Common metric variation

The gravity and matter sectors are varied with respect to the same source-derived metric. In the long-wavelength, low-curvature, vanishing-hypermomentum sector, the source-native first-order gravity functional takes the metric-affine form

$$S_{\text{IR}}[g, \Gamma, \lambda] = \frac{1}{2\kappa_B} \int d^4x \sqrt{-g} g^{\mu\nu} R_{(\mu\nu)}(\Gamma) + S_m[g, \lambda], \quad \kappa_B = \frac{3}{5}. \quad (49)$$

The value $3/5$ is the normalized outer-cup routing ratio derived before the metric variation. It is not selected by matching an Einstein target. The compatible connection equation removes the projective mode and gives $\Gamma = \Gamma_{\text{LC}}(g)$ in this sector. Metric variation therefore gives

$$G_{\mu\nu}(g) = \frac{3}{5} T_{\mu\nu}(g, \lambda). \quad (50)$$

The Bianchi identity is consistent with Theorem 11.1; neither side is assigned a conservation law independently to force agreement.

Theorem 12.1 (Scoped noncircular Einstein backreaction). *In the fixed pointed-Braid model and the continuum class of Theorem 11.1, the source-native first-order gravity sector and full-corner matter sector share the same metric variation. Their Euler–Lagrange equation is Eq. (50). No new Einstein–Hilbert term, mixed fitting term, manual $3/5$, or independent stress/Ward axiom is introduced at this stage.*

This is the precise sense in which the previous conditional SDPC bridge is removed in v0.8. It is not a theorem over every imaginable Braid model or every continuum action. It is an internally derived result in one fixed source model and one explicitly delimited universality class.

12.2 Low-energy spin two

Linearize $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ about the flat source anchor. The quadratic gravity action belongs to the massless Fierz–Pauli gauge class [4],

$$\mathcal{L}_{\text{FP}} = -\frac{1}{2} \partial_\lambda h_{\mu\nu} \partial^\lambda h^{\mu\nu} + \partial_\mu h^{\mu\nu} \partial^\lambda h_{\lambda\nu} - \partial_\mu h^{\mu\nu} \partial_\nu h + \frac{1}{2} \partial_\lambda h \partial^\lambda h, \quad (51)$$

with gauge symmetry $\delta h_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$. Coupling to the conserved tensor (46) preserves the linear gauge constraint. In four dimensions the physical massless sector has two helicities.

The claim is a low-energy sector statement. It does not quantize the interacting continuum metric, establish a graviton Fock space on arbitrary backgrounds, or prove perturbative or nonperturbative ultraviolet completion.

12.3 Finite-to-continuum residual

Let $h = 5^{-n}$ be the declared refinement scale. The first variation of the finite action converges weakly to Theorem 12.1 with the proved bound

$$G_{\mu\nu} - \frac{3}{5} T_{\mu\nu} = O(h). \quad (52)$$

The gravity contribution is bounded by $O(h^2)$, while the current matter regularity proof limits the combined estimate to $O(h)$. Each five-adic refinement therefore decreases the stated upper bound by at least a factor $1/5$.

The actual event involution already pairs every directed edge with its reverse. The local matter functional is consequently the Jeffreys pair

$$D(\rho_x \| \rho_y) + D(\rho_y \| \rho_x). \quad (53)$$

All odd terms in its midpoint Bregman expansion, including the cubic term, cancel exactly. This is not a post hoc symmetrization. It removes the orientation-asymmetry coefficient but does not improve the global bound (52) without stronger inter-cell regularity. An explicit analytic positive-rate counterexample with three log rates $(0, 0, 1)$ shows that internal Markov analyticity alone does not match derivatives across adjacent spacetime cells.

13 Full finite quantum evolution and its present boundary

13.1 Reflection-positivity obstruction

The unmodified full finite Umegaki/Jeffreys functional does not satisfy the tested standard positive one-link transfer condition. Inside the actual 125-dimensional source charge family, the fixed packet $\mu = (-1, 2, 6)$ at source coupling $\alpha = 1/50$ gives a cross-link 3×3 determinant

$$\det K_{\text{link}} = -0.008747721561 \dots < 0. \quad (54)$$

This is a direct reflection-positivity NO-GO for that transfer route, not a disproof of the infrared theorem. A BKM heat kernel exists on the local-label manifold but is not a full spacetime-slice transfer matrix. The involution-doubled block is positive semidefinite, yet has 25 zero modes in dimension 50 and is an idempotent projector rather than finite Hamiltonian evolution.

13.2 Source-native GKSL evolution

The fixed source independently supplies a completely positive trace-preserving semigroup of Gorini–Kossakowski–Sudarshan–Lindblad type [7, 9]. Fixing the event clock rate (24) does not determine the three remaining Hamiltonian controls; distinct generators share the same clock. More importantly, trace preservation is probability conservation, not the gravitational Ward law. An open-system balance must include the environment charge.

Let $\{U_g\}_{g=1}^6$ be the six real Braid actions in the Reynolds channel and let

$$p = \left(\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8} \right). \quad (55)$$

The isometry

$$V\psi = \sum_{g=1}^6 \sqrt{p_g} U_g \psi \otimes |g\rangle_E \quad (56)$$

gives an exact Stinespring dilation at each collision [12]. A finite fixed environment cannot reproduce the entire exponential relaxation autonomously for all time. The native Braid right tail solves the capacity problem: each three-strand block decomposes as

$$125 = 17 \cdot 1 + 28 \cdot 3 + 4 \cdot 6, \quad (57)$$

so it contains four regular six-state registers, and the infinite tail supplies fresh collision ancillas.

The pointed and neutral tail states do not select a regular copy or basepoint. This apparent selector is gauge for the reduced channel. Across eight source sectors and four copies, all marked blocks are related by diagonal-sign environment unitaries that preserve the relevant projector. Changing copy or basepoint changes V only by an environment unitary and leaves the partial-trace channel invariant.

13.3 Exact temporal total balance

The system-only G_1 defect has exact Hilbert–Schmidt norm squared 18 in every sector. It is not a total nonconservation. There exists a six-state environment charge H_E such that

$$V^\dagger (G_1 \otimes I_E + I \otimes H_E) V = G_1 \quad (58)$$

in all eight sectors. The equation is covariant under the environment torsor gauge. For N fresh collisions it telescopes to

$$V_N^\dagger \left(G_1^{(S)} + \sum_{k=1}^N H_{E,k} \right) V_N = G_1^{(S)}|_{t=0}. \quad (59)$$

Thus the dissipative temporal defect is exactly a charge transfer into the Braid tail.

13.4 Spatial range NO-GO

For the three tested spatial source candidates

$$H_i = -i[G_1, P_{\chi_i}], \quad i = 1, 2, 3, \quad (60)$$

the channel defects are nonzero in every sector, with squared norms

$$(24, 14, 14). \quad (61)$$

They are exactly orthogonal to the entire six-state environment-only compression range

$$\text{span}\{U_g^\dagger U_h : g, h = 1, \dots, 6\}. \quad (62)$$

No environment-only charge can cancel them. The temporal balance (58) remains exact, but a four-component discrete Ward identity requires either source-derived system–environment interaction stress or a different Noether momentum. At the BGCE311 boundary this was the shortest finite-quantum completion gate.

The statement above is the exact BGCE311 boundary retained from v0.8. It is not the final v0.81 result. The next construction preserves the environment-only NO-GO while deriving the missing term as a source-record variation and a history-dressed interaction current.

14 Finite interaction stress and the local recorded Lorentz parent

14.1 Raw/Petz branch contrast

Let the four source scores be

$$S_\alpha \in \{G_1, H_1, H_2, H_3\}, \quad D_\alpha = (I - \Phi)S_\alpha, \quad \alpha = 0, 1, 2, 3, \quad (63)$$

where Φ is the source Reynolds collision channel at $q = e^{-\pi}$. The source modular reference fixes a raw/Petz $\mathbb{Z}_2$ record grading. It is distinct from the forward/backward doubling of a closed-time-path contour.

For the common two-sided positive normalization of the conditional branches, the conditional effect tangents at the base point satisfy

$$\dot{F}_{+, \alpha} = \frac{1}{2}D_\alpha, \quad \dot{F}_{-, \alpha} = -\frac{1}{2}D_\alpha. \quad (64)$$

Consequently,

$$\dot{F}_{+, \alpha} - \dot{F}_{-, \alpha} = D_\alpha, \quad \frac{1}{2}(\dot{F}_{+, \alpha} + \dot{F}_{-, \alpha}) = 0. \quad (65)$$

The second equality explains why the prior-summed canonical Stinespring connection vanishes: the source response is antisymmetric in the two record branches. It is not absent.

For branch Choi operators $J_{\pm, \alpha}$,

$$\text{Tr}_{\text{out}}(J_{+, \alpha} - J_{-, \alpha}) = D_\alpha^\top. \quad (66)$$

This marginal is invariant under arbitrary unitary Kraus mixing and phases within each fixed record branch. Mixing raw and Petz is not admitted because it erases the source-fixed modular grading rather than changing a Kraus representation.

The local trajectory score

$$\mathcal{A}_\alpha(t; \sigma) = \text{Tr} \sigma [\log F_{+, \alpha}(t) - \log F_{-, \alpha}(t)] \quad (67)$$

therefore obeys

$$\left. \frac{d}{dt} \mathcal{A}_\alpha(t; \sigma) \right|_{t=0} = \text{Tr}(\sigma D_\alpha). \quad (68)$$

Thus D_α is the finite Hilbert representative of the metric variation. The temporal D_0 is the already derived environment transfer. The three spatial D_i are the nonzero interaction defects, whose physical-support lifts are uniquely $I_i = VD_iV^\dagger$ in the declared support class. No cancelling operator is chosen after the defect is inspected.

14.2 Cellwise CTP construction and refinement

At source-cylinder stage m , take

$$\mathcal{A}_m = C(C_m) \otimes M_{125}. \quad (69)$$

The central address algebra $C(C_m)$ labels operational cells, while M_{125} carries the collision fibre. Every forward and backward contour copy contains the same raw/Petz-recorded local instrument. The fixed prior sum is CTP, and the record contrast retains Eq. (64). The two doublings are not identified.

Cellwise direct sum, spatial tensor product, source-event permutation, and causal composition preserve complete positivity and trace preservation. They also preserve diagonal CTP normalization, branch Hermiticity, largest-time cancellation, source-retarded support, and positive-semidefinite noise. The source event form

$$K_{\text{evt}} = \frac{1}{25} \begin{pmatrix} 36 & -16 & -16 & -16 \\ -16 & 36 & -16 & -16 \\ -16 & -16 & 36 & -16 \\ -16 & -16 & -16 & 36 \end{pmatrix} \quad (70)$$

has eigenvalues $(-12/25, 52/25, 52/25, 52/25)$ and supplies the $1+3$ signature without a manual Wick sign.

Address refinement, tail refinement, and five-way event-time refinement satisfy

$$\mathcal{L}_{m+1}[jg]j = j\mathcal{L}_m[g], \quad q_{m+1}^5 = q_m, \quad 625 \cdot 625^{-(m+1)} = 625^{-m}. \quad (71)$$

The result is therefore a source-derived local Lorentzian recorded CTP parent in the aligned four-score, block-local finite-process class. It is not yet a full ten-component nonlinear metric parent or a theory with nontrivial spatial-gradient self-interactions.

15 Four-component total Ward identity

For each source score, the strong one-collision intertwiner has the form

$$(Y_\alpha^{(0)} + J_\alpha)V = VX_\alpha, \quad V^\dagger J_\alpha V = D_\alpha. \quad (72)$$

The same D_α is both the branch-score metric derivative and the interaction-current compression. Hence the Ward insertion is not reverse-engineered from a desired conserved equation.

Later collisions dress the insertion along causal descendants. On the reduced algebra,

$$\sum_{k=0}^{n-1} \Phi^k(D_\alpha) = S_\alpha - \Phi^n(S_\alpha). \quad (73)$$

For $\Phi = E + qQ$ and $D_\alpha \in Q\mathcal{A}$,

$$\Phi^k(D_\alpha) = q^k D_\alpha, \quad \sum_{k=0}^{n-1} \Phi^k(D_\alpha) = \frac{1 - q^n}{1 - q} D_\alpha. \quad (74)$$

Equation (73) holds for all four components, all eight actual source sectors, and every tested finite history. It is natural under five-way time refinement and 625-child address refinement. In the refinement limit it gives a weak Ward statement for compatible compactly supported cylinder test fields.

This does not reinstate the excluded environment-only spatial current. The current is interaction-bearing and history dressed. Nor does it prove an arbitrary smooth diffeomorphism Ward identity or the hypersurface-deformation algebra of continuum general relativity.

16 A finite matter–geometry–matter cycle

16.1 Record Fisher action

The recorded collision supplies a score matrix M of rank four in every source sector and every source-selected control order. For record word x , define

$$S_x = \frac{1}{2}M^\top(x - p_\omega \mathbf{1}), \quad F = \frac{1}{4}M^\top M, \quad (75)$$

and the strictly convex action

$$\Gamma_x(\delta h) = \frac{1}{2}\delta h^\top F \delta h - S_x^\top \delta h. \quad (76)$$

Its Euler equation and unique stationary update are

$$F\delta h = S_x, \quad \delta h_x = F^{-1}S_x. \quad (77)$$

The output and input metric registers carry the same aligned source-cylinder tangent type. Closing them by the identity map introduces no coefficient or new operator. Together with the global positive logistic retraction, one complete cycle is

$$\rho_n \mapsto x_n, \quad h_{n+1} = h_n + F_n^{-1}S_n(x_n), \quad \rho_{n+1} = \Phi_{h_{n+1}}(\rho_n \mid x_n). \quad (78)$$

The first arrow is a joint CP instrument, the second is a deterministic classical-register translation, and the last is a metric-controlled CPTP channel. Their composition is CP and the record sum is TP. All 768 sector/order/clock paths satisfy the Euler equation, and all 48 expected updates are nonzero in all four components.

Equation (78) is a direct, noncircular finite backreaction construction in its declared class: the target Einstein tensor, Einstein–Hilbert and Fierz–Pauli actions, MMR, CGR, manual 3/5, and coefficient fitting are absent. Its limitation is equally explicit. The identity closure is the minimal type-preserving rule, but the source does not prove that no other endomorphism defines a physically admissible dynamics.

16.2 All-finite-iteration nonlinear closure

BGCE457 closes the repeated-update question inside the aligned four-score, source-cylinder, operator-logistic, identity-feedback family. Let the finite metric register be h and let

$$D(h) = \sum_{a=0}^3 h^a D_a, \quad F_{\text{raw}}(h) = I + \tanh\left(\frac{D(h)}{2}\right), \quad F_{\text{Petz}}(h) = I - \tanh\left(\frac{D(h)}{2}\right). \quad (79)$$

Both effects are strictly positive at every finite h , and their fixed-prior average is exactly the identity. Each current-register record instrument, deterministic register translation, and next metric-controlled collision is therefore CPTP after record summation. Induction gives a CPTP composition for every finite iteration count n .

The Ward identity also survives a feedback-dependent channel sequence. For $P_0 = \mathbf{1}$ and $P_{k+1} = P_k \circ \Phi_k$,

$$\sum_{k=0}^{n-1} P_k[S_a - \Phi_k(S_a)] = S_a - P_n(S_a). \quad (80)$$

This is an exact nonautonomous telescoping identity rather than an asymptotic conservation fit. Moreover,

$$\log F_{\text{raw}}(h) - \log F_{\text{Petz}}(h) = D(h), \quad dD_h(\delta h) = \sum_a \delta h^a D_a, \quad (81)$$

so the four global response directions are independent at every finite register. Their Hilbert–Schmidt Gram rank is four in all eight actual source sectors, with recorded minimum eigenvalue 13.019585510459214. The operator $\tanh$ makes the next record law genuinely nonlinear at finite amplitude. Because feedback changes the conjugation-natural instrument parameter but not the source algebra or its generated derivations, it introduces no new central term into the scoped constraint representation.

This proves all-finite-iteration closure, not uniqueness among all dynamics, an $n \rightarrow \infty$ uniform bound, an off-base pathwise Fisher lower bound, standard graviton scattering, or continuum strong-curvature quantum gravity.

16.3 The coefficients 1 and 3/5 have different referents

If Eq. (76) is written with a relative source coefficient,

$$\Gamma_{\kappa,x}(\delta h) = \frac{1}{2} \delta h^\top F \delta h - \kappa S_x^\top \delta h, \quad (82)$$

the same-parent score and Fisher Hessian select

$$\kappa_{\text{direct}} = 1. \quad (83)$$

This value is invariant under invertible metric-coordinate changes, record replication, and common action rescaling. By contrast, the earlier value

$$\kappa_{\text{cup}} = \frac{3}{5} \quad (84)$$

is the normalized full-corner outer-cup routing ratio. Applying 3/5 to the whole Fisher action changes no stationary equation; applying it only to the score leg gives the exact relative Euler residual 2/5. A natural same-parent comparison map transports the first and second variations together and therefore preserves coefficient 1. Obtaining 3/5 as the direct Fisher coupling would require an asymmetric score/Hessian map with ratio $b/a = 3/5$, which is absent from the fixed source packet.

The v0.8 continuum theorem is not withdrawn: it still uses the independently derived full-corner routing ratio in its declared infrared action class. Version 0.81 instead records that the direct finite Fisher coupling and the continuum routing coefficient have not been identified as the same physical quantity. A complete microscopic-to-Einstein derivation must either derive a typed asymmetric bridge or explain why the two coefficients belong to different observables.

17 Calibration, matter, and constraint boundaries

17.1 Absolute units

The finite theory is dimensionless. Independent rescalings of the internal clock and stress units leave every collision record and the relative coefficient 1 unchanged. Source-only seconds, metres, energy, and Newton coupling are therefore nonidentifiable. In the declared two-scale class, one independent frequency anchor and one independent stress or gravity anchor are sufficient for a locally unique calibration map. Version 0.92 retains the exploratory device-relative AWS frequency result and CODATA G as an external input, but the provider-clock SI traceability and full uncertainty budget remain unavailable.

17.2 Current matter sector

The source-derived internal algebra in all eight sectors is

$$\mathfrak{h} \cong \mathfrak{sl}(3, \mathbb{R})^2 \oplus \mathfrak{sl}(2, \mathbb{R})^3. \quad (85)$$

It is not the compact Standard Model algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$. In particular, compact $\mathfrak{su}(3)$ cannot embed into the current direct sum: a nonzero projection to a simple factor would have to be injective, which is excluded either by target dimension or by the inequivalent split and compact real forms. The current module also lacks source-typed chiral complex representations, hypercharge, anomaly data, a Higgs sector, and Yukawa interactions. This is an exact mismatch for the present source, not a no-go for every future source extension.

17.3 Finite gauge closure versus gravitational constraints

The 25-generator internal matrix Lie algebra and its compact dual close exactly under commutators, satisfy the Jacobi identity, and persist under every passive finite refinement $X \mapsto X \otimes I_5$. This is anomaly-free finite internal algebraic closure. It is not a gravitational first-class constraint algebra. The source packet has one clock constraint $p_t + H_B = 0$ but not an Einstein Hamiltonian constraint, three spatial momentum constraints, their full bracket table, or a continuum hypersurface-deformation algebra. Deriving or excluding a source clock plus three spatial moment-map constraints is therefore the next internal completion gate.

18 Constraint-algebra closure through BGCE443

BGCE377–442 tested the shortest candidate routes from the exact internal closure to a physical gravitational constraint system. Several algebraic constructions close formally, but closure alone is not sufficient: the generators must act on the physical metric and feedback variables, possess a source-fixed symplectic or presymplectic meaning, and reproduce a refinement-compatible first-class bracket rather than a relabelled internal commutator algebra.

The search excluded, within their declared classes, direct promotion of the internal Lie generators, projector and incidence variants, homogeneous normalizer routes, and arbitrary coherent Stinespring-phase geometry. The last point is especially restrictive. A fixed quantum instrument admits coherent dilations whose Berry curvature can have different spatial ranks, including zero and two. Consequently the instrument does not identify an arbitrary dilation phase as physical spacetime curvature. This is a source-identifiability failure, not merely a missing numerical fit.

Three later gauge-invariant routes also fail. BGCE392 finds zero null-tetrad-selected mean Uhlmann spatial rank in all eight sectors. BGCE441 finds zero spatial rank for the support-restricted raw/Petz relative-logarithmic commutator form. BGCE442 reconstructs the fixed finite source algebra as a 34-dimensional unital transpose-closed real algebra with center dimension nine and

$$\dim \text{Der}(\mathcal{A}) = \dim \text{Inn}(\mathcal{A}) = 25, \quad HH^1(\mathcal{A}, \mathcal{A}) = 0. \quad (86)$$

Its separability also removes positive-degree Hochschild-cohomology rescue by changing only the coefficient bimodule. Thus the fixed finite source algebra has no outer three-plane. BGCE443 therefore moves to the pointed refinement limit rather than trying to manufacture outer directions at one finite level.

From the exact eight-sector source snapshot, the three signed spatial generators have rank three in every sector. Constructive nonuniform product-state witnesses give exact gap $2/3$ in all 24 signed directions. Hence the order- n canonical finite-range refinement has spectral-diameter and inner-derivation-norm lower bound

$$\|\delta_i^{(n)}\| \geq \frac{2}{3}(n-2), \quad n \geq 3, \quad (87)$$

so the limit cannot collapse to a bounded inner derivation. Refinement compatibility defines three closable unbounded derivations on the dense local algebra. For each of the seven nonzero signed sectors, all three raw local-generator actions differ from the unsigned reference, giving 21 exact Braid-specificity witnesses.

At each finite order, define the constraint algebra as the Lie algebra generated by these three derivations. Matrix commutators close by construction, the Jacobi identity vanishes exactly in all eight sectors, and scalar central terms disappear in the derivation representation. Thus the source-generated local-derivation constraint algebra is anomaly-free in this declared scope.

The claim ceiling is essential. The three seed generators need not themselves form a three-dimensional Lie algebra; finite-range closure may contain further generators. No cotangent moment-map identification, Hamiltonian-versus-momentum typing, ADM algebra, or hypersurface-deformation algebra is derived. Version 0.92 therefore closes the registered source-generated constraint-algebra task, not every conventional formulation of gravitational constraints.

19 Scoped ultraviolet and all-scale closure through BGCE456

The ultraviolet question separates into at least two non-equivalent requirements: existence of quantum dynamics at every source-refinement depth and control of nonlinear metric composite observables. BGCE445 closes the first requirement in the declared block-local parent. At stage m the observable algebra is a finite direct sum of matrix algebras,

$$\mathcal{A}_m = C(C_m) \otimes M_{125}, \quad (88)$$

and the child-constant refinement $j_m : \mathcal{A}_m \rightarrow \mathcal{A}_{m+1}$ is an injective unital $*$ -homomorphism. The compatible finite-stage CPTP channels have UCP Heisenberg adjoints, hence extend uniquely to the AF/quasi-local completion $\mathcal{A}_\infty$. For every event time t ,

$$\|\Phi_{\infty,t}\|_{cb} = 1. \quad (89)$$

The finite-stage operator-norm time limit and the common complete bound imply a strongly continuous limit semigroup on the dense local algebra. The local Ward telescope is refinement-natural and survives there. This establishes an all-refinement quantum-channel completion without counterterms, cutoff-dependent coefficients, or a target Einstein equation. It does not by itself bound composite stress, the limit generator on every domain, or continuum spatial gradients.

BGCE444–449 close a second, narrower nonlinear response problem. For a four-dimensional carrier,

$$\dim \text{Sym}^2(V_4^*) = 10, \quad \dim \text{Sym}^4(V_4^*) = 35, \quad (90)$$

and

$$\dim \text{Sym}^2(\text{Sym}^2(V_4^*)) = 55 = 35 + 20. \quad (91)$$

The source solder $S_\chi : A_{10} \rightarrow \text{Sym}^2(V_4^*)$ has rank ten in every sector. Total symmetrization of its quadratic response therefore gives

$$D^2Q_\chi : \text{Sym}^2(A_{10}) \longrightarrow \text{Sym}^4(V_4^*), \quad \text{rank } D^2Q_\chi = 35, \quad \dim \ker D^2Q_\chi = 20. \quad (92)$$

This is a source-fixed capacity theorem rather than a fitted lift of a known target.

The finite source cylinder also supplies an exact fourth-jet rule. One refinement has five digits on each of four axes, hence $5^4 = 625$ children. The degree-zero-through-four Vandermonde matrix on nodes 0, 1, 2, 3, 4 has rank five; its fourfold tensor product reconstructs every fourth jet exactly on the degree-at-most-four cylinder-polynomial subspace. There are 70 total-order-at-most-four scalar jet entries and 700 for the ten metric fields. The rule is S_4 -equivariant and commutes with the rank-ten solder. It is not a uniform approximation theorem for arbitrary nonpolynomial fields.

BGCE447 then identifies the actual UB612 PBM metric with the same ordered four-carrier. The source frame

$$\left(G_1, \frac{\{G_1, P_{\chi_1}\}}{2}, \frac{\{G_1, P_{\chi_2}\}}{2}, \frac{\{G_1, P_{\chi_3}\}}{2} \right) \quad (93)$$

is exactly the ordered $G_1, X_{\chi_1}, X_{\chi_2}, X_{\chi_3}$ frame used by the PBM lineage. The Hadamard frame, unit-triangular clock orthogonalization, positive creation scale, and PBM coordinate transport compose to a fixed invertible map. It commutes with metric jets because it is coordinate-independent and linear; this is not an arbitrary post-hoc $GL(4)$ alignment.

Finally, BGCE448 constructs all 55 unordered second-response columns of the actual 35-component UB612 tensor U_4 . BGCE449 selects a source-fixed 35-column minor and proves its nonsingularity throughout the complete saved componentwise interval box. A 220-digit inverse is used only to propose a preconditioner; the final Krawczyk–Beeck–Collatz bound is evaluated in exact rational arithmetic:

$$\rho_{\text{cert}} \leq 0.012750158721617564 < 1. \quad (94)$$

Consequently every admissible 35×55 response matrix in that box has row rank 35. The earlier apparent last-mode instability was a conditioning and column-selection problem, not loss of a physical quartic mode.

BGCE450–455 then separate ultraviolet damping from geometric propagation. Raw/Petz refinement alone cannot supply a physical ultraviolet cutoff. A source spatial Hodge Laplacian gives positive unbounded spatial dynamics but fails child-constant refinement. The corrected hierarchical source Laplacian

$$\Delta_m = \sum_{k=1}^m 25^k D_k \quad (95)$$

intertwines refinement exactly, generates UCP heat, has spectral dimension three, and gives finite polynomial heat moments at every $t > 0$. Metric-volume weighting supplies a finite Hilbert stress and metric Ward identity on the child-constant hierarchy. The associated Lorentzian wave has hierarchical dynamic exponent $z = 1$.

The raw inverse-metric exterior operator preserves a positive wave but fails the Markov sign condition. BGCE456 removes this obstruction using the same saved actual inverse spatial metric $A = \gamma^{-1}$. On the source child group C_5^3 , let $\delta(x, y) \in \{-2, -1, 0, 1, 2\}^3$ be the unique centered lift of $y - x$ and define

$$c_A(x, y) = \frac{1}{5} \frac{\delta(x, y)^T A \delta(x, y)}{\delta(x, y)^T \delta(x, y)}, \quad x \neq y. \quad (96)$$

All 7,750 unordered edge conductances are strictly positive. For $A = I$, $c_I = 1/5$ and the child Laplacian is exactly $25(I - E_0)$, fixing the normalization without fitting. Axis conductances remain unequal for the actual metric, so anisotropy is not erased.

Placing this same child Laplacian inside every level- k parent block gives

$$\Delta_m^A = \sum_{k=1}^m 25^{k-1} L_A^{(k)}, \quad \Delta_{m+1}^A j_m = j_m \Delta_m^A. \quad (97)$$

Because L_A is a symmetric nonnegative-conductance graph Laplacian, $e^{-t\Delta_m^A}$ is positivity preserving, unital, trace preserving, and UCP after the internal matrix lift. The same positive self-adjoint operator generates the wave equation $\partial_s^2 \phi + \Delta_m^A \phi = 0$. Exact form bounds $\alpha \Delta_{\text{iso}} \leq \Delta_A \leq \beta \Delta_{\text{iso}}$ imply spatial spectral dimension three by min–max comparison, while level energies scale by 25 and wave frequencies by five.

The package status is therefore *closed scoped*: the all-refinement block-local quantum channel, actual UB612 same-carrier quartic response, and actual-metric anisotropic hierarchical UCP heat/positive-wave carrier are established. It is not a full physical UV completion. The conductance map is not proved unique from source axioms; active nonpolynomial PBM volume projectivity, equality to an exact smooth principal symbol, the full metric Fréchet Hessian, arbitrary nonpolynomial source sampling, the complete σ_6 /parallel operator, uniform composite-stress control, and smooth-spacetime renormalization remain outside the proof.

20 Derived Standard-Model matter sector

20.1 Unchanged-source boundary and derived extension

BGCE375 remains valid: the unchanged split-real internal algebra is not Standard-Model equivalent. Version 0.82 does not erase that result by renaming its generators. Instead, BGCE393–439 asks whether additional structure already selected by the source’s complete anomaly-solution class can define a controlled derived extension.

The construction starts from the eight source-selected $\{D, U\}$ anomaly orientations and three intrinsic predicate flips. They form

$$X = (\mathbb{Z}_2)^3, \quad G = (\mathbb{Z}_2)^3, \quad \mathcal{A}_{\text{ASG}} = C(X) \rtimes G \cong M_8(\mathbb{C}). \quad (98)$$

The 64 operators $E_x T_g$ give every matrix unit exactly once, and the regular representation is faithful. This groupoid is declared as a derived extension. It is not an operation already present in the unchanged event star, time reversal, raw/Petz pair, or CTP parent; BGCE438 proved that those existing operations do not generate the required cube translations.

20.2 Three degree-one chiral copies

Because the three cube coordinates are source-defined predicates, Boolean degree measures response order to those predicates. The degree multiplicities are

$$(1, 3, 3, 1). \quad (99)$$

The degree-one projector therefore has rank three and commutes with the gauge action common to all cube vertices. Each of its three directions carries the same left-Weyl character

$$(3, 2)_1 \oplus (\bar{3}, 1)_{-4} \oplus (\bar{3}, 1)_2 \oplus (1, 2)_{-3} \oplus (1, 1)_6. \quad (100)$$

Thus the degree-one module contains three identical copies of the Standard-Model chiral character. The associated global gauge group is

$$\frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}. \quad (101)$$

For all eight source sectors, the local gauge and mixed anomalies vanish, the weak Witten parity is even, and the global kernel remains $\mathbb{Z}_6$.

The mere fact that a Boolean cube has a rank-three first Fourier shell would not establish three physical generations. The additional source-specific facts are that this cube is the complete minimum-anomaly solution class, its coordinates are intrinsic source predicates, and every vertex independently carries the same derived chiral character. Even so, identifying linear anomaly-solution response with physical generations remains an explicit scoped interpretation rather than a measured fact.

20.3 One independent Higgs doublet and Yukawa types

The degree-zero scalar sector has orientation rank one. On the conjugate routes $(1, 2)_3 \oplus (1, 2)_{-3}$, a gauge-equivariant anti-linear involution $J_H = AK$ satisfies $J_H^2 = 1$. Its fixed real form has real dimension four. Hence the independent scalar content is one complex weak doublet; the conjugate doublet is dependent, not a second freely specifiable Higgs field.

The source cubic

$$\mathcal{J}(A, B, C) = \text{Tr}(ABC) \quad (102)$$

contains the up-type, down-type, and charged-lepton Yukawa interaction cycles. Groupoid averaging retains generation-diagonal couplings between equal degree-one characters and gives rank three without coefficient fitting.

20.4 Claim ceiling

This closes representation and interaction *types* only in the explicitly derived groupoid and Boolean-response class. It does not derive observed Yukawa values, fermion masses, neutrino content, CKM or PMNS mixing, electroweak vacuum dynamics, confinement, running couplings, or empirical Standard-Model confirmation. Nor is the groupoid yet forced as the unique physical extension of the finite gravity parent. Those are separate dynamical and empirical gates.

21 Executed calibration and metrological boundary

21.1 Source-only nonidentifiability remains exact

The source-only absolute-calibration NO-GO of BGCE374 is unchanged: dimensionless collision data do not determine seconds, metres, stress units, or Newton's constant. The positive clock/stress scale orbit has dimension two. External anchors can select a representative, but they do not turn absolute units into a Braid-only theorem.

21.2 E2-audited device-relative frequency anchor

Before outcome access, the source spectrum $\{0, 6, 10\}$ fixed the unique minimum positive calibration gap $\Delta h_{\text{cal}} = 4$ and transition $6 \leftrightarrow 10$. BGCE440 Revision 19 then completed a frozen signed-quadrature Ramsey packet on Rigetti Cepheus-1-108Q through Amazon Braket: twelve programs, 64 shots each, and 768 retained per-shot outcomes. The independently replayed endpoint passed all ten noncompensating gates. Its evidence class remains exploratory.

The fitted implemented-Braid transition frequency is

$$f_{\text{cal}} = 4,484,999,494.698898 \text{ Hz}, \quad u(f_{\text{cal}}) = 9,314.910565190186 \text{ Hz}, \quad (103)$$

$$R^2 = 0.9825714110512933. \quad (104)$$

The signed detuning is 1,274,614.64466693 Hz and the fitted amplitude is 0.4282978085915778. With the frozen conversion

$$\tau_* = \frac{\Delta h_{\text{cal}}}{2\pi f_{\text{cal}}}, \quad \ell_* = c\tau_*, \quad (105)$$

the registered device-relative units are

$$\tau_* = 1.419442238778493 \times 10^{-10} \text{ s}, \quad (106)$$

$$\ell_* = 4.255380777524273 \times 10^{-2} \text{ m}, \quad E_* = 7.429480314095178 \times 10^{-25} \text{ J}. \quad (107)$$

Independent E2 audit exactly replayed the estimator from the retained raw results, reproduced every numerical endpoint and gate, and matched all twelve public AWS task records. This validates the result revision and its device-relative interpretation; it does not make the result confirmatory, universal, or SI-traceable.

21.3 CODATA-input conditional stress scale

Revision 20 uses the official 2022 CODATA recommended value

$$G = 6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2} \quad (108)$$

as the second independent scalar anchor, with the SI-exact $c = 299,792,458 \text{ m s}^{-1}$ [29, 30]. The direct-parent coupling remains frozen at $\kappa = 1$; no gravity response, target Einstein coefficient, MMR/CGR bridge,

manual 3/5, or coefficient fit is used. The conversion is

$$\kappa_{\text{cal}} = \frac{8\pi G}{c^4} = 2.076647442844972 \times 10^{-43} \text{ m/J}, \quad (109)$$

and

$$\sigma_* = \frac{\kappa}{\kappa_{\text{cal}}(c\tau_*)^2} = 2.659257381834094 \times 10^{45} \text{ Pa}. \quad (110)$$

Propagating only the CODATA uncertainty and the Revision-19 statistical fit gives

$$u_r(\sigma_*) = 2.285490726338230 \times 10^{-5}, \quad u(\sigma_*) = 6.077708085128302 \times 10^{40} \text{ Pa}. \quad (111)$$

This is a *partial* uncertainty. Because G is an external input, the result is a conditional unit calibration, not a derivation or prediction of Newton's constant and not an independent test of gravity.

21.4 Public-metadata traceability NO-GO

Revision 21 audits the live Cepheus capability schema, its versioned native calibration payload, and official Amazon Braket documentation [31, 32]. The 13,722 scalar capability fields and the native calibration record contain no clock, oscillator, traceability, uncertainty, accuracy, stability, Allan-deviation, ppm, or ppb field. The public documents confirm periodic provider calibration, but do not publish an SI-reference chain or numerical timebase uncertainty.

This is a route-specific NO-GO: public user-visible metadata cannot complete the SI traceability proof. It is not evidence that the hardware is uncalibrated. Closing the calibration now requires a device- and execution-applicable Rigetti or AWS metrology attestation identifying the reference timebase, certificate or traceability chain, validity interval, and accuracy/stability uncertainty. Until then, τ_* , ℓ_* , E_* , and σ_* remain device-relative or conditionally calibrated quantities, and BQG-G3-R03.2 remains externally pending.

22 Braid-first reconstruction of finite quantum structure

Version 0.92 asks a question logically prior to the gravity limit: how much finite quantum structure can be reconstructed without assuming a complex Hilbert space, a Born rule, or a full observable algebra? The answer is a scoped reconstruction on the actual three-history source carrier.

22.1 Oriented history cycle and complex carrier

Let R_1 and R_2 be the two adjacent actions induced by the actual 25-event pointed-Braid table on 125 three-history atoms, and let

$$C = R_1 R_2, \quad A = C - C^{-1}, \quad 3P = 2I - C - C^{-1}. \quad (112)$$

The involutive Yang–Baxter identity gives $C^3 = I$ exactly. Direct evaluation on all 125 atoms gives

$$A^2 = -3P. \quad (113)$$

The permutation C has 17 fixed points and 36 three-cycles. Consequently P has real rank 72, and

$$J = \frac{A}{\sqrt{3}}, \quad J^2 = -I \quad \text{on } \mathcal{H} = \text{im } P, \quad (114)$$

turns $\mathcal{H}$ into a complex vector space of dimension 36. Even words commute with J ; the odd adjacent generators reverse its sign and therefore act as conjugation-type maps. No complex matrix, quantum state, phase fit, or Born probability is used as input.

This mechanism is not uniquely Braid-specific. The identity control gives complex dimension zero, a deterministic Yang–Baxter-broken control fails $C^3 = I$, but the ordinary flip gives complex dimension 40. The result is therefore an exact construction from the actual pointed-Braid source, not a theorem that only pointed braids can generate complex kinematics.

22.2 Source dagger and squared-norm probability

Source reversal supplies the dagger, while equality counting and cup/cap closure supply the positive pairing. For a nonzero ray $\psi \in \mathcal{H}$ and a source-typed self-adjoint projector E , the probability is forced within this declared linear gluing class to be

$$p(E \mid \psi) = \frac{\langle E\psi, E\psi \rangle_{\text{src}}}{\langle \psi, \psi \rangle_{\text{src}}}. \quad (115)$$

For the first two canonical three-cycles and the witness $\psi = x + 2y$, the projector onto the first cycle gives $p = 2/10 = 1/5$. Power rules with exponents one and four remain positive, normalized, coordinate-symmetric, phase-blind, and product-multiplicative for this witness, but fail the parallelogram identity. The quadratic rule alone agrees with the source dagger closure.

Equation (115) is not presented as a new general derivation of the Born rule. Squared norms from linear dagger-compatible closure are standard quantum and categorical mathematics. The new claim is the internal source route that constructs the required finite complex carrier, dagger, projectors, and counting pairing in one recorded model.

22.3 Connectivity, superselection, and the observable algebra

The initial local words and diagonal source-cylinder projectors preserve 32 active S_3 orbit blocks. Even after overestimating every block by its full matrix algebra, their complex algebra dimension is only 44 rather than $36^2 = 1296$. Thus naive universal connectivity is exactly impossible.

The source cup contains an endpoint exchange $1 \leftrightarrow 3$. Its three history-site placements reduce the 32 support components to 11. Adding the fixed source Brauer crossings and caps reduces them to two. These components are not an accidental missing link. They are the two eigenspaces of the unique nontrivial binary grading class

$$b(0) = b(2) = b(4) = 0, \quad b(1) = b(3) = 1, \quad \chi(a, b, c) = (-1)^{b(a)+b(b)+b(c)}. \quad (116)$$

The grading commutes with the source transition, endpoint exchange, all eight signed-sector Brauer crossings/caps, and the complex projector. The 36 complex cycle lines split as $18 + 18$. Inside each sector, rank-one source-cylinder projectors and the connected off-diagonal morphisms generate every matrix unit. Therefore

$$\mathcal{A}_{\text{obs}} = M_{18}(\mathbb{C}) \oplus M_{18}(\mathbb{C}), \quad \mathcal{A}'_{\text{obs}} = \mathbb{C} \oplus \mathbb{C}. \quad (117)$$

This closes complex kinematics, finite projective probability, the superselection charge, and the full observable algebra within each source sector. It does not by itself promote every observable to an independently drivable physical Hamiltonian.

23 Finite stochastic quantum engine and actuation boundary

23.1 Compressed controls and the mixed-unitary engine

The source already selects four raw controls G_1, H_1, H_2, H_3 , the internal duration $t = \pi/3$, and a uniform law on the six orders of the three spatial controls. The raw 125-dimensional controls do not preserve the

36-dimensional complex carrier. The source counting/cup pairing nevertheless fixes a canonical orthogonal compression. With B the normalized $+i$ carrier basis, define

$$O_{\text{eff}} = B^* O B. \quad (118)$$

For every signed sector and both $\mathbb{Z}_2$ blocks, the four compressed controls are Hermitian, preserve the grading, and are noncommuting. The six ordered pulses U_σ are projectively distinct. Hence

$$\mathcal{E}(\rho) = \frac{1}{6} \sum_{\sigma \in S_3} U_\sigma \rho U_\sigma^* \quad (119)$$

is CPTP, unital, and has Kraus/Choi rank six on all 16 blocks. This is a finite mathematical quantum engine, not yet a bare-source hardware circuit.

23.2 Finite carrier control and exact route exclusions

The C_3 Reynolds map

$$\mathcal{R}_C(O) = \frac{1}{3} \sum_{k=0}^2 C^k O C^{-k} \quad (120)$$

reproduces the canonical compression on the carrier. A qutrit dilation using controlled I, C, C^2 implements the associated twirl exactly and minimally. Exhaustion of all 720 six-strand Coxeter word classes proves, however, that no finite word in the old adjacent Braid generators realizes this controlled action. One marked generator

$$U(x, k) = (C^k x, k), \quad k \in \mathbb{Z}_3, \quad (121)$$

is necessary and sufficient in the declared action-groupoid extension. Unoriented coherence leaves the pair $k \mapsto \pm k$; matching controller orientation to the already selected positive orientation of C chooses $k \mapsto k$.

The same generator yields the exact carrier projector

$$P_\omega = \frac{I + \omega^2 C + \omega C^2}{3}, \quad S = I - 2P_\omega. \quad (122)$$

Nevertheless, a two-segment reflection echo fails all 32 sector/control cases, because the carrier-preserving and leakage pieces do not commute. A one-shot twirl channel also does not equal the exponential of the averaged Hamiltonian. These NO-GO results prevent the finite twirl from being misreported as the full engine.

23.3 Finite spectral synthesis, source-word LCU, and signal clock

Each compressed control H is a finite Hermitian matrix, so its distinct spectral values λ_j determine an exact polynomial representative

$$p_H(H) = e^{-i\pi H/3}, \quad \deg p_H \leq 35. \quad (123)$$

Across the 32 cases there are 15–32 distinct spectral nodes and interpolation degrees 14–31. This exact identity is not yet a circuit. Indeed, a finite alphabet of permutation, root-of-unity, rational-counting, and algebraic-square-root gates cannot produce the required transcendental phase for an algebraic irrational eigenvalue; the obstruction follows from the Gelfond–Schneider theorem.

The source controls nevertheless admit coefficient-free LCU decompositions. In particular,

$$\frac{G_1}{5} = \frac{15I - 7r_1 - 4r_2 - 4r_1 r_2 r_1}{30}, \quad (124)$$

and each $H_i/10$ is a 32-branch LCU built from the same four words, a four-character mismatch projector, and the two commutator orders. A single typed coherent source-word operation

$$\text{SELECT}_W = \sum_j |j\rangle\langle j| \otimes e^{i\phi_j} W_j \quad (125)$$

implements all 32 block encodings exactly in the declared extension. This is one generator type, but it is broader than the old controlled- C word and is not derived as a finite word in the original adjacent generators.

The previously obtained normalized four-probability carrier now has an exact operational role. Its V_4 character matrix maps the uniform direction and the three sum-zero contrasts to

$$(G_1, H_1, H_2, H_3). \quad (126)$$

Thus the 1 + 3 probability decomposition is an outer control selector, not merely a dimension match. It does not by itself preserve the common/standard decomposition after the four different operations are coherently selected.

An independent raw-source clock copy supplies the continuous signal resource. The raw G_1 spectrum is $\{0, 2, 3, 5\}$ in every signed sector. Its rank-36 eigenspaces at 2 and 3 define a logical signal qubit tensored with a 36-fold degeneracy. On this subspace,

$$G_1 = \frac{5}{2}I + \frac{1}{2}Z, \quad r_2 = \frac{1}{2}Z + \frac{\sqrt{3}}{2}X, \quad XZ + ZX = 0. \quad (127)$$

The source flow e^{-itG_1} and conjugation by r_2 generate continuous $SU(2) \otimes I_{36}$ without selecting individual degenerate eigenvectors.

23.4 Exact finite actuation and its efficiency NO-GO

LCU, block encoding, and quantum signal processing are standard algorithmic primitives [33, 34, 35]. Their use here does not constitute invention of those primitives. The source-specific result is the exact decomposition, spectrum, typed selector, and signal-clock realization for the fixed Braid engine.

For each distinct eigenvalue, define the Lagrange projector

$$L_j(H) = \prod_{k \neq j} \frac{H - \lambda_k I}{\lambda_j - \lambda_k}. \quad (128)$$

Every affine factor is exactly block encoded from the source-word LCU and identity. With

$$\beta_j = \frac{\prod_{k \neq j} (\alpha + |\lambda_k|)}{|\prod_{k \neq j} (\lambda_j - \lambda_k)|}, \quad s = \sum_j \beta_j, \quad (129)$$

an outer preparation with weights β_j/s and phase $\text{sgn}(D_j)e^{-i\pi\lambda_j/3}$ produces the exact success block

$$\frac{1}{s} e^{-i\pi H/3}. \quad (130)$$

The success probability is input-independent and equals $1/s^2$. It ranges from approximately $10^{-45.89}$ to $10^{-88.47}$. Since the success amplitude is known, arbitrary-phase amplitude amplification can rotate it exactly to one [36, 37]. The exact deterministic circuit, however, requires approximately $10^{23.14}$ to $10^{44.43}$ calls to the heralded actuator. A Bernstein-polynomial lower bound gives the same $\Omega(s)$ black-box scaling. Therefore deterministic existence is closed, while practical black-box implementation is ruled out for the current normalization.

This NO-GO is retained: it applies to amplification of the Lagrange actuator as a fixed black box. It does not exclude a prior change of polynomial basis that lowers the normalization itself.

23.5 Low-normalization Chebyshev compiler and source-word walk

BGCE490 performs that change before amplification. Put $A = H/\alpha$ with $\|A\| \leq 1$ and let

$$B_E(A) = \begin{pmatrix} A & \sqrt{I - A^2} \\ \sqrt{I - A^2} & -A \end{pmatrix}, \quad R = \text{diag}(I, -I)B_E(A). \quad (131)$$

Because A and its defect belong to the same functional calculus,

$$\langle 0 | R^k | 0 \rangle = T_k(A). \quad (132)$$

The unique finite-spectrum interpolant can therefore be written

$$P_H(x) = \sum_{k=0}^{m-1} c_k T_k(x), \quad P_H(\lambda_j/\alpha) = e^{-i\pi\lambda_j/3}. \quad (133)$$

Across all 32 sector/control cases the coefficient one-norm satisfies $0.5304 \leq \log_{10} \sum_k |c_k| \leq 3.2497$. The unamplified success probability is approximately $10^{-6.4994}$ to $10^{-1.0609}$, an improvement of 39.40–87.10 decimal orders over the Lagrange construction. Exact phase matching then gives unit success with 7–2,793 base calls and 203–39,102 H -block-encoding queries. The maximum 70-digit interpolation and walk residuals are respectively 3.13×10^{-69} and 1.82×10^{-70} .

BGCE491 removes the apparent square-root-defect primitive. Let U_A be the existing source-word LCU unitary with Hermitian success block A , and define its one-qubit hermitianization

$$\tilde{U}_A = \begin{pmatrix} 0 & U_A \\ U_A^\dagger & 0 \end{pmatrix}. \quad (134)$$

If E embeds the encoded signal state and LCU success address, $E^\dagger \tilde{U}_A E = A$. With $S_E = 2EE^\dagger - I$ and $W = S_E \tilde{U}_A$, the recurrence $B_{k+1} = 2AB_k - B_{k-1}$ proves

$$E^\dagger W^k E = T_k(A) \quad (135)$$

exactly. All eight signed sectors, four controls, spectral nodes, and degrees through 31 pass; the largest numerical Chebyshev residual is 1.14×10^{-14} . Thus no explicit $\sqrt{I - A^2}$ or spectral-read primitive is required, and the 203–39,102 query range is retained.

The remaining boundary is now physical rather than exponential. The construction uses source-counted prep, the typed source-word select, source reversal/dagger, the encoded signal qubit, and a typed success-address reflection. The last address mark and enlarged finite-list controller have not been derived from bare adjacent-Braid relations or compiled to a calibrated hardware gate set. No quantum computer has executed this engine, and no empirical quantum-gravity claim follows from the finite circuit theorem.

24 Physical compilation and exploratory real-QPU execution

Version 0.91 ended with an exact source-word Chebyshev walk at the abstract query level. The post-v0.91 compiler sequence asks a narrower operational question: can a finite source action be represented on a current gate-model quantum processor without changing its exact source action, and can its prespecified functional probes survive hardware noise?

The first compiled realization used 102 logical CNOT gates and 330 native CZ gates. A reversible label/center/side coordinate change reduced the exact source action to two coordinate swaps and one center-XOR-controlled Fredkin, giving a 46-CNOT logical circuit. Physical placement on a seven-qubit Cepheus

patch reduced this to 73 CZ gates. A subsequent phase analysis found a source-arrow $\mathbb{Z}_4$ cocycle: the 34-CNOT representative is not equal to the canonical representative in raw computational phase coordinates, but it is gauge-conjugate to it on the valid source states and satisfies the Yang–Baxter equation, including accumulated phase, on all 125 valid three-source atoms. The local observable algebra and Born pairings are covariant under that phase frame.

The original functional decoders were not phase-frame invariant. A bounded zero-CNOT retarget search selected a single output S phase on qubit q_5 . The resulting circuit retained 34 logical CNOT gates and compiled on the frozen patch to 61 CZ gates and two-qubit depth 45. BGCE531 then executed PLUS, MINUS, BASIS, and ordinary-swap probes as four sequential 256-shot tasks on Rigetti Cepheus. All tasks completed without retry for a settled cost of USD 1.6352. The frozen sample-level gates were

$$p_0^+ = 0.13872832, \quad p_0^- = 0.18681319, \quad p_0^{\text{basis}} = 0.15028902, \quad p_0^{\text{swap}} = 0.06640625, \quad (136)$$

with source-workspace retention between 0.6758 and 0.7109. The BASIS all-zero output was the unique sample mode, and $|p_0^+ - p_0^-| = 0.04808486 \leq 0.20$. BGCE531 is therefore a scoped exploratory real-QPU functional-compatibility pass.

That statement is deliberately weaker than statistical certification. BGCE533 re-analysed the frozen counts with exact familywise tests. Workspace retention remained decisive after Holm correction, and the PLUS and MINUS zero counts each exceeded the ordinary-swap comparator with one-sided Fisher probabilities 0.0104146 and 0.000108484. The BASIS lead, however, was only 26 counts against a largest rival of 23. Its exact conditional tail was 0.3877248, far above the 63-rival Bonferroni threshold 0.00079365. Thus the sample-level decision remains historically valid, but the claim that all-zero is the population BASIS mode is not statistically certified.

Later controls sharpened the interpretation. Restricting the BASIS endpoint to the independently defined 25 valid two-source states leaves all other strings as explicit leakage and makes all-zero the unique valid-state sample mode, but neither the retained sample nor a fresh 998-shot Braid task met the prespecified familywise certificate. An exactly resource-matched ordinary-swap control instead produced its prespecified nonzero valid state $(2, 4)$ as a statistically certified mode in a fresh 998-shot task. This excludes the simplest explanation that valid-state projection or device zero bias alone forces the Braid all-zero mode. The Braid and control tasks had identical published calibration snapshots and equal gate counts, but different physical edge-use distributions. A same-skeleton angle-only control has therefore been constructed but not yet executed.

These results establish three things and no more:

1. the phase-dressed finite source generator has an exact 34-CNOT representation in the declared source phase frame;
2. one frozen functional packet passed at sample level on a real QPU, while its strongest BASIS population claim did not pass exact finite-sample certification;
3. matched controls disfavor projection-only and simple zero-bias explanations, while edge-specific coherent error and general Braid necessity remain open.

The full 203–39,102-query Chebyshev actuator was not run. The experiment does not establish quantum advantage, a gravity interaction, quantum-gravity validation, or a uniquely Braid explanation of the observed counts.

25 All-real-time strong convergence on the hierarchical two-polarization carrier

The earlier finite-to-continuum theorem controlled first variations by an $O(h)$ residual but did not supply strong trajectory convergence. BQGEULER-004 closes a sharper theorem for the linear wave sector already generated by the BGCE452/BGCE456 hierarchy.

Let H_m be the mean-zero finite-level carrier, $J_m : H_m \rightarrow H_{m+1}$ the child-copy isometry, and $\Delta_m \geq 0$ the self-adjoint spatial operator. The source hierarchy supplies the exact relation

$$\Delta_{m+1}J_m = J_m\Delta_m. \quad (137)$$

Introduce two passive polarization slots,

$$H_m^{(2)} = H_m \otimes \mathbb{R}^2, \quad \Delta_m^{(2)} = \Delta_m \otimes I_2, \quad (138)$$

and let $\mathcal{E} = D(\Delta^{1/2}) \otimes \mathbb{R}^2 \oplus H \otimes \mathbb{R}^2$ be the inductive-limit energy space. The first-order wave system is

$$\dot{q} = p, \quad \dot{p} = -\Delta q. \quad (139)$$

Exact intertwining and functional calculus make every finite-level wave group $W_m(t)$ a restriction of one limit group $W(t)$. If Q_m is the energy projection onto level m , unitarity gives the exact identity

$$\sup_{t \in \mathbb{R}} \|W(t)x - J_m W_m(t) Q_m x\|_{\mathcal{E}} = \|(I - Q_m)x\|_{\mathcal{E}} \longrightarrow 0. \quad (140)$$

This is uniform for all real time, not merely on every compact interval. For the retained two-polarization witness with level- k amplitudes $q_k = 25^{-k}$ and $p_k = 5^{-k}$, the squared isotropic energy tail is exactly

$$\frac{1}{6}25^{-m}. \quad (141)$$

The actual-metric positive-conductance operator obeys exact positive form bounds relative to the isotropic operator and the same refinement intertwining, so the theorem transfers to the retained anisotropic hierarchical carrier.

The theorem closes linear two-slot strong convergence on the source hierarchy, but not yet the physical graviton statement. The two copies of $\mathbb{R}$ have the correct mode count, while the source has not supplied a refinement-compatible physical ADM projector P_m satisfying

$$P_{m+1}J_m = J_m P_m, \quad P_m \Delta_m = \Delta_m P_m, \quad (142)$$

nor an exact identification of the hierarchical principal symbol with the smooth ADM/Fierz–Pauli symbol. Consequently Eq. (140) is not a proof of nonlinear Einstein evolution, physical ADM convergence, or a smooth-spacetime continuum limit. It removes a genuine stability gap while isolating the remaining source-identification problem.

26 Higgs, Yukawa, fermion, flavour, and neutrino update

The v0.91 matter section derived representation content and Yukawa *types*; it did not derive a Higgs vacuum, Yukawa values, or fermion masses. The post-v0.91 sequence adds a finite source-relative spectrum while preserving the distinction between representation theory, action selection, calibration, and observed-particle identification.

26.1 Generation operator and nonzero Higgs minimum

The pointed source selects a rank-six module equal to three copies of the real two-dimensional standard representation. On its three-dimensional multiplicity space, closure of the source marker algebra first gives a positive generation operator with primitive spectrum $(5, 4, 2)$. A stronger pointed right-difference KMS response subsequently gives a full-rank nonnormal operator Y whose three singular values are distinct. Exact Sturm certificates yield the unfitted relative generation spectrum

$$1 : 7.6620663145 : 24.6714340497. \quad (143)$$

The source-central KMS metric also fixes unequal relative squared species strengths

$$a_u : a_d : a_e = 1 : \frac{315}{356} : \frac{135}{178}. \quad (144)$$

No observed quark or lepton mass was used to choose either ratio.

The source sign line and matter parity select the ordinary exterior/Fock trace rather than the parity-inserted supertrace. With $A(H) = |H|^2 M^* M \geq 0$, finite matter marginalization gives

$$S_{\text{eff}}(H) = S_H(H) - \log \det(I + A(H)). \quad (145)$$

For the later variational selector, the resulting finite potential is strictly convex in the retained radial variable and has a unique positive stationary point. Its source-normalized radius is approximately

$$|H| = 0.0031298399315334307. \quad (146)$$

This supersedes the earlier provisional radius only in the later selector scope. It is a dimensionless finite-source vacuum coordinate, not the measured electroweak vacuum expectation value.

The pointed affine first-order KMS response action supplies a variational bridge: the exact source variation $W(1) - W(0)$ equals Y , the forward and reverse CTP trilinears carry Y and $Y^\top$, and the mixed Higgs variation produces the same positive $Y^\top Y$ entering Eq. (145). This removes an arbitrary relative Yukawa matrix inside that declared class.

26.2 Device-relative spectra and their boundaries

Combining Eqs. (143) and (144) produces nine positive nondegenerate dimensionless bare mass labels. A pointed-oriented trine functor synchronizes the matter and source-clock Lie frames and fixes the previously free dimensionless conversion constant to $c = 1$ in that class. The exploratory device anchor then converts the table to energy equivalents ranging from approximately 4.19×10^{-12} to 1.19×10^{-10} eV.

These are not measured fermion masses. They are device-relative spectral labels. Public provider metadata do not close the SI traceability chain; the source does not identify the three intrinsic ascending eigenvalues with electron/muon/tau or particular quark generations; and the retained static parent does not select a unique RG trajectory.

A clock-typed retarded resolvent and source-trace effective Feshbach projection nevertheless add a finite interacting layer. In the declared pointed-relative subtraction class, the CTP point eigenvalue uniquely selects $\Sigma_{\text{ren}}(z) = \Sigma(z) - \Sigma(0)$. The resulting positive finite parent has three positive subthreshold poles per species. These poles are source-derived finite dressed spectral outputs in that class, not Standard-Model pole masses or a renormalization-group flow. Only the inherited statistical device-frequency error propagates; provider systematic, universal-scale, RG/model, and observed-label uncertainties remain unavailable rather than zero.

26.3 Flavour and neutrino boundary

For flavour, conjugate Higgs routes and the forward/reverse CTP branches select $Y_u = Y$ and $Y_d = Y^\top$ in the combined derived branch-chirality class. Both left- and right-mass commutators have exact rank two, excluding simultaneous diagonalization and identity quark mixing within that class. A later complementary triangular-support condition fixes a relative coefficient and sign pattern in its declared endpoint. The physical CKM and PMNS matrices, CP phase, observed-generation dictionary, and experimental comparison remain open.

The minimal BGCE439 left-Weyl character contains no gauge-singlet $(1, 1)_0$ carrier, so neutrino mass does not close in the unchanged matter source. A declared unital matter-functor completion promotes the canonical algebra-unit line tensored with the rank-three generation shell to exactly three neutral odd carriers while

preserving local anomalies, Witten parity, and the global $\mathbb{Z}_6$ kernel. Normalized Frobenius structure alone leaves phase/sign controls; after adding unitality and the inherited real-positive convention, the dimensionless Dirac coefficient is uniquely

$$y_\nu = +1 \quad (147)$$

inside that explicit completion. The unchanged source does not itself prove that the physical neutrino Yukawa trilinear is this unital endomorphism, so this is a conditional scoped pass, not an unconditional Braid-only coefficient theorem.

The relative Majorana pair $K = YY^\top$ is fixed, but its dimensionful action coefficient is not. The exact positive assignments

$$M_R = E_{\text{clock}}, \quad M_R = 2E_{\text{clock}} \quad (148)$$

preserve clock synchronization $c = 1$, the normalized dagger-compact pair, zig-zag normalization, gauge neutrality, and parity while changing the Majorana action. Thus clock rate and pair normalization do not derive a Majorana mass scale. A further theorem would require a source-native joint clock-pair action and mixed variation; none is assumed in this version.

27 Why subjectivity intersection is operative

The construction is not obtained from braid relations alone. It uses a pointed event effect, nonidentical typed source roles, retained transport provenance, a marked causal clock, and a rule for which intersection output is public. These data do physical work:

1. the event effect selects the operational corner;
2. deletion or collapse of a standpoint changes the source packet;
3. the marked event involution supplies the neighbor law and reverse-edge pairing;
4. the clock line selects the causal reflection and its co-moving extension;
5. the source-product typing makes the ten-component solder unique;
6. the right tail supplies an endogenous environment for reduced quantum evolution.

An unpointed abstract braid group does not provide these roles. An alternative theory could append equivalent marks, states, and provenance; doing so would recreate the operative subjectivity-intersection structure under different terminology.

The philosophical interpretation is not forced uniquely by the algebra. The same structure may be described as marked relational quantum events or provenance-preserving quantum reference data. The SIEL interpretation is that a standpoint is a typed mode of access and that objectivity is what remains stable under lawful intersection and re-description. Nothing in this statement attributes consciousness to a braid, source sector, clock, or quantum channel.

28 Constructive refutation of objectivist primacy

The Objectivist Primacy Thesis OP is universal: it says that every stable, shareable, transformation-respecting carrier must be primitive or equivalent to a primitive perspective-neutral carrier. The present construction has the order

$$\begin{aligned} &\text{nonidentical pointed source standpoints} \\ &\quad \longrightarrow \text{marked event, matter corner, and quantum channel} \\ &\quad \longrightarrow \text{four null directions, causal reflection, and Lorentz carrier} \\ &\quad \longrightarrow \text{glued tensor, conserved stress, and scoped backreaction.} \end{aligned} \quad (149)$$

The public carrier is not present as a completed background at the first step. It is reconstructed, chart-glued, and dynamically coupled while retaining the provenance that generated it. Identity and flip comparators fail essential rank properties, while deletion of the pointing removes the selections used by the construction.

Therefore OP is false as a necessity thesis: at least one exact model generates a stable Lorentzian public carrier from nonidentical standpoint-bearing source processes. This conclusion does not deny objective regularity or mind-independent existence. It denies only that ontological priority of a completed objective carrier is universally necessary. The stronger claim that nature realizes this construction remains empirical and is not established here.

29 How complete is the quantum-gravity theory?

The theory now satisfies several requirements that a mere analogy does not:

1. a finite noncommutative quantum source with unitary Braid transport and CPTP channels;
2. a source-derived four-dimensional Lorentz carrier and local tensor field;
3. matter and geometry read from one marked source rather than independently normalized models;
4. a selected long-wavelength action, Hilbert stress, and on-shell Ward identity in a declared class;
5. noncircular Einstein backreaction and massless spin-2 recovery in that class;
6. a finite open-system dilation with exact temporal total-charge transfer;
7. a local raw/Petz-recorded Lorentzian CTP parent with four metric-response directions;
8. a four-component strong-vertex and history-dressed total Ward identity;
9. all-finite-iteration, parameter-free, CPTP-compatible nonlinear matter–geometry–matter feedback in the operator-logistic identity-feedback class;
10. a Braid-specific rank-three closable unbounded local-derivation constraint algebra with exact finite-order Lie/Jacobi closure;
11. a derived anomaly-solution groupoid with three anomaly-free Standard-Model chiral character copies, the $\mathbb{Z}_6$ global quotient, one independent complex weak doublet, and source-cubic Yukawa interaction types;
12. an E2-audited exploratory device-relative frequency anchor and a conditionally populated two-anchor stress-scale map.
13. a norm-one AF/quasi-local quantum channel at all refinement depths, an exact source-fixed rank-35 response for the actual UB612 quartic tensor throughout its saved interval box, and an actual-metric anisotropic hierarchical UCP heat/positive-wave carrier of spectral dimension three.
14. a finite Braid-history reconstruction of complex kinematics, source-dagger Born probabilities, an exact $\mathbb{Z}_2$ superselection charge, and the observable algebra $M_{18}(\mathbb{C}) \oplus M_{18}(\mathbb{C})$;
15. a source-selected six-order mixed-unitary quantum engine, an exact finite heralded actuator, the retained Lagrange black-box efficiency NO-GO, and an exact low-normalization Chebyshev replacement using 203–39,102 block-encoding queries.
16. a 34-CNOT source generator and an exploratory real-QPU functional PASS, together with exact finite-sample and matched-control boundaries;
17. all-real-time, energy-norm strong convergence for two polarization copies of the mean-zero hierarchical source-cylinder wave carrier;
18. a finite source-relative Higgs/Yukawa/fermion spectrum with a variational nonnormal generation operator, a nonzero source-normalized Higgs minimum, and source-selected finite dressed poles in the declared classes.

It is not a completed quantum-gravity theory because the following remain open:

- extension of the finite parent from the aligned four-score, block-local class to full ten-component non-linear metric dynamics with spatial gradients;
- source necessity or uniqueness of the identity-feedback law, and uniform control of nonlinear adaptive backreaction as iteration number tends to infinity;

- a physical bridge from the all-real-time hierarchical two-slot wave theorem to a refinement-compatible ADM quotient and the exact smooth Fierz–Pauli principal symbol;
- quantization of the continuum gravitational degrees of freedom beyond the finite source and low-energy quadratic sector;
- extension of the scoped all-refinement channel, quartic-response theorem, and anisotropic conductance carrier to a full metric Fréchet Hessian, arbitrary nonpolynomial fields, bounded composite stress, an exact smooth principal symbol, and physical smooth-spacetime UV completion;
- cotangent moment-map and Hamiltonian/momentum typing of the source-generated constraints, including comparison with the ADM hypersurface-deformation algebra and a continuum diffeomorphism limit;
- a complete SI-traceable uncertainty budget and a Braid-predicted, rather than externally supplied, dimensionful gravitational coupling;
- a source-forced observed-particle realization of the derived matter groupoid, including a universal scale, physical generation names, CKM/PMNS values, CP phase, RG flow, total uncertainty, and empirical mass matching;
- unconditional typing of the neutrino extension and a source-native joint clock-pair action that fixes the dimensionful Majorana scale;
- cosmological/black-hole solutions and distinct calibrated numerical predictions;
- derivation or minimality proof of the typed source-word select, finite-list controller, signal interface, and success-address mark relative to the bare adjacent-Braid vocabulary;
- compilation and execution of the full 203–39,102-query Chebyshev actuator, beyond the executed 34-CNOT phase-frame source generator;
- a gravity-specific observable residual for an atom interferometer, optical clock, torsion balance, or other apparatus that differs prospectively from general relativity plus ordinary quantum matter;
- independent replication, peer review, and natural-data validation.

The appropriate description is therefore *a source-explicit mathematical Braid quantum-gravity theory candidate whose finite quantum structure, low-normalization finite actuation, registered internal constraint, all-finite backreaction, and anisotropic UCP gates are closed in declared scopes, with a separately scoped Einstein limit and derived Standard-Model representation sector*. Calling it empirically established, bare-Braid hardware complete, or an unrestricted continuum completion would be premature; calling it only a suggestive analogy would ignore the exact quantum, tensor, parent, Ward, constraint, iterative-backreaction, all-scale-channel, anomaly, and representation theorems.

30 Falsifiability and next gates

30.1 Formal falsifiers

The exact package fails under revision-matched replay if any of the following occurs:

- the event involution loses its Yang–Baxter relation, rank-two pair moment, or reverse-edge pairing;
- the event average depends on hidden physical axis weights;
- a nontrivial declared partition preserves the eleven-operator matter packet;
- the source-cup routing fails to yield 3/5 without retuning;
- the null links lose rank four or K_{cvt} loses Lorentz signature;
- the refinement clock ceases to fix $2\pi/3$ in the declared completion;
- the rational carrier identities, tensor gluing, horizontal flatness, or typed solder fail exact regeneration;
- the co-moving map loses rank ten or develops an independent clock coupling;
- the finite variation does not converge to the Hilbert variation in the stated class;
- the gravity and matter variations require different metrics or a fitted mixed term;
- the environment gauge changes the reduced channel or invalidates Eq. (58);
- the branch contrast fails to reproduce any D_α or loses within-branch Kraus invariance;

- the local parent loses CPTP normalization, causal CTP identities, or refinement naturality;
- the history-dressed Ward telescope (73) fails in any component or sector;
- the Fisher matrix in Eq. (76) loses rank, or the finite update ceases to define a CPTP-controlled next collision.
- any finite operator-logistic iteration loses CPTP normalization, violates Eq. (80), or reduces the global four-response Gram rank;
- any BGCE443 source sector loses rank three, any constructive direction loses its exact 2/3 gap, or the generated finite-order derivation algebra develops a nonzero Jacobi residual or central anomaly;
- the anomaly-solution transformations fail to generate $M_8(\mathbb{C})$ exactly, the degree-one projector loses rank three or gauge equivariance, or any stated local/global anomaly test fails;
- the Higgs real form has dimension other than four, or the source cubic lacks any of the three stated Yukawa interaction cycles.
- replay of the retained 768 Ramsey shots fails any frozen endpoint gate or does not reproduce the stated frequency and device-relative conversions;
- the CODATA-input stress conversion changes under the frozen formulas, or is represented as a Braid prediction of G rather than an external-input calibration.
- the finite-stage channels fail refinement compatibility, the AF-limit UCP extension loses complete bound one, or the local Ward observable fails refinement naturality;
- the rank-ten source solder fails to induce rank 35 on the quartic response, the actual UB612 same-carrier intertwiner becomes singular, or the exact BGCE449 interval upper bound is not strictly below one.
- any actual-metric conductance in Eq. (96) is nonpositive, the isotropic limit differs from $25(I - E_0)$, refinement intertwining fails, or the anisotropic spectral counting exponent differs from three.
- the actual three-history cycle fails $C^3 = I$, Eq. (115) fails the source dagger closure, the declared binary grading fails to commute with any retained source morphism, or Eq. (117) loses sectorwise irreducibility;
- any of the 32 compressed controls loses Hermiticity or grading preservation, the six-order channel in Eq. (119) ceases to be CPTP/unital of Kraus rank six, or the exact spectral pulse in Eq. (123) fails revision-matched replay;
- the heralded LCU success block is not exactly $s^{-1}e^{-i\pi H/3}$, its probability depends on the input state, or the phase-matched amplification fails to produce exact unit success probability under the declared arithmetic model;
- Eq. (132) or Eq. (135) fails, any of the 32 Chebyshev interpolants misses a declared spectral node, or the reproduced query count leaves the certified 203–39,102 range.
- the 34-CNOT phase-frame circuit fails exact source-state action, phase-dressed Yang–Baxter closure, or the reported QPU counts fail replay from the retained task artifacts;
- Eq. (137) or Eq. (140) fails on the retained hierarchy, or the exact witness energy tail differs from $(1/6)25^{-m}$;
- the declared Higgs determinant loses its unique positive finite minimum, the variational response ceases to equal the retained Y , or the exact generation/species spectra fail regeneration;
- the conditional neutrino carrier changes any stated anomaly or parity check, or the two retained Majorana-scale countermodels cease to preserve the pinned clock and dagger-compact data.

30.2 Decisive next theorems

The interaction-stress, four-component Ward, source-generated constraint-algebra, all-finite-iteration back-reaction, anisotropic hierarchical UCP, and finite source-dagger quantum-reconstruction gates are now closed in their declared classes. BQGEULER-004 additionally closes all-real-time strong trajectory convergence for the hierarchical two-slot wave sector. The immediate continuum theorem is no longer another stability estimate; it is a source-derived refinement pullback for the physical ADM projector and an exact smooth Fierz–Pauli principal-symbol identification. Nonlinear Euler convergence remains a later gate.

For the finite quantum engine, BGCE490–491 close the low-normalization exact compiler and eliminate the explicit square-root-defect and spectral-read primitives. The later compiler sequence reaches a real-QPU

phase-frame source-generator run, but does not execute the full actuator. The next empirical compiler gate is the already constructed same-physical-skeleton angle-only control, followed by a statistically powered replication only if prospectively authorized. A positive control separation would still not be quantum advantage or gravity evidence.

For matter, relative Higgs/Yukawa structure and a finite source/device-relative spectrum are now available, but observed-particle identification is not. The shortest remaining neutrino theorem is a source-native joint clock-pair action whose mixed variation selects a dimensionful Majorana coefficient. The two exact positive countermodels in the current clock/pair data prohibit setting that coefficient by convention. In parallel, flavour work must derive a physical observed-label dictionary and numerical CKM/PMNS output without fitting.

The decisive programme gate remains empirical. BGCE440 Revision 19 is a real-hardware exploratory calibration outcome, not a gravity discriminator. Q-012/X1004 supplies a complementary-basis quantum prediction candidate, but its carrier-to-instrument map and standard-quantum competitor set remain unresolved. For gravity, no existing gravimeter, atom interferometer, or optical clock can yet adjudicate the theory because no frozen Braid-versus-GR residual has been derived. The shortest experimental target is therefore a dimensionless differential observable in which general relativity plus ordinary quantum matter predicts zero while a source orientation or intersection structure predicts a fixed nonzero phase, redshift, or acceleration residual. This is a research target, not a v0.92 result. No gravitational response, natural-data confirmation, or empirical Standard-Model confirmation is claimed.

31 Claim ledger

Statement	Status	Exact scope
Operational event cylinder and formal CGR retirement	Exact derived	Fixed marked source
Full matter corner and rank-ten response	Exact derived	Fixed event-conditioned corner
MMR2 routing	Closed	Complete five-partition retraction class; not arbitrary instruments
Relative factor 3/5	Source-derived	Normalized outer-cup routing; not a calibrated Newton prediction
Rank-four event moment Q_4	Exact derived	24-chart source average
Four-null-link Lorentz carrier	Exact derived	Fixed source; no universal dimension theorem
IR causal action tensor	Exact scoped theorem	Real Stone/affine completion and source reflection semigroup
CPTP quantum channel on causal carrier	Exact derived	Eight ordered-edge Gram sectors
Finite $GL(4)$ transport/nonmetricity	Exact derived	48 source-cube packets
Continuous metric/nonmetricity tensor	Exact scoped theorem	Smooth source cylinder; passive local $GL(4)$
Active finite Markov transport	Exact local theorem	BKM-horizontal exponential leaf
Physical metric solder	Unique in class	Source-product-generated Sym^2 typing
Hilbert stress and Ward identity	Final scoped pass	Long-wavelength, local, second-order, self-adjoint, conservative class
Einstein equation $G = (3/5)T$	Final scoped pass	Same class; low curvature and zero hypermomentum

Statement	Status	Exact scope
Massless spin-2 with two helicities	Derived IR output	Flat four-dimensional quadratic limit
Finite-to-continuum residual	$O(h)$ bound	Odd midpoint terms cancel; $O(h^2)$ not proved globally
Full finite one-link reflection positivity	NO-GO	Unmodified tested Umegaki/Jeffreys kernel
Source-native GKSL evolution	Exact derived	Reduced finite quantum semigroup
Temporal total G_1 balance	Exact derived	Six-state Braid tail environment; all eight sectors
Spatial environment-only balances	NO-GO	Three tested commutator momenta
Raw/Petz branch interaction stress	Exact scoped theorem	Four aligned scores; one-collision physical support
Local recorded Lorentzian CTP parent	Final scoped pass	Aligned four-score, block-local source cylinder
Four-component finite total Ward	Final scoped pass	Strong vertex and causal history-dressed reduced current
All-finite-iteration backreaction	Final scoped pass	Operator-logistic identity-feedback class; every finite composition CPTP, exact Ward telescope, response rank four; no uniqueness or uniform nonlinear infinite-iteration theorem
Direct finite relative coefficient	Exact scoped value 1	Same-parent score/Hessian action
1 versus 3/5 relation	Closed scoped by typing	Internal same-record Fisher gain is 1; outer matter/gravity source-cup routing weight is 3/5; they are distinct quantities, not competing estimates
Source-only absolute calibration	NO-GO	Two independent external anchors required; dimensionless scale orbit remains exact
Unchanged-source Standard Model equivalence	Exact scoped NO-GO	Split-real internal algebra; existing source operations do not generate the anomaly cube
Derived anomaly-solution groupoid	Final scoped pass	$C((\mathbb{Z}_2)^3) \rtimes (\mathbb{Z}_2)^3 \cong M_8(\mathbb{C})$
Three chiral matter copies	Exact scoped derivation	Degree-one Boolean response; three identical anomaly-free SM characters
Global gauge group	Exact scoped derivation	$[SU(3) \times SU(2) \times U(1)]/\mathbb{Z}_6$
Higgs representation	Exact scoped derivation	One independent complex weak doublet via gauge-equivariant real form
Yukawa interaction types	Exact scoped derivation	Up, down, and charged-lepton source-cubic cycles
Variational relative Yukawa operator	Closed scoped	Pointed affine first-order KMS response action; no observed-mass fit
Nonzero finite Higgs minimum	Closed scoped	Unique source-normalized minimum for the later variational selector; not measured electroweak scale

Statement	Status	Exact scope
Relative fermion spectrum	Closed scoped	Nine positive nondegenerate dimensionless bare labels and finite dressed poles in declared classes
Absolute/observed fermion masses	Open	Device-relative only; universal scale, label dictionary, RG flow, total uncertainty, and empirical match absent
Flavour misalignment	Partial scoped pass	Exact noncommuting up/down singular bases in a declared branch-chirality class; numerical CKM/PMNS and CP phase open
Neutrino carrier and Dirac coefficient	Conditional scoped pass	Three neutral carriers and $y_\nu = +1$ in a unital Frobenius-positive completion; unchanged-source physical typing open
Dimensionful Majorana scale	Scoped NO-GO for present clock/pair route	One- and two-clock-unit positive countermodels preserve pinned structure; joint source action required
Finite internal Lie anomaly	Absent in scope	Exact Jacobi closure and passive refinement
Source-generated constraint algebra	Closed scoped	Braid-specific closable unbounded local derivations with exact finite-order Lie/Jacobi closure
ADM/hypersurface-deformation identification	Open	No cotangent moment-map or Hamiltonian/momentum typing theorem
AF/quasi-local all-refinement channel	Closed scoped	Compatible UCP limit with cb-norm one and refinement-natural local Ward
Source fourth-jet reconstruction	Exact scoped theorem	$5^4 = 625$ child tensor rule; degree-at-most-four polynomial subspace
Actual UB612 same-carrier map	Exact derived	Fixed invertible source-to-PBM carrier map; no post-hoc $GL(4)$ fit
Actual 35×55 quartic response	Closed scoped	Rank 35 over the full saved interval box; exact bound $0.012750158721617564 < 1$
Actual-metric anisotropic UCP carrier	Closed scoped	Positive conductance hierarchy, exact refinement, same-operator UCP heat/positive wave, spectral dimension three
Two-slot hierarchical wave convergence	Closed scoped	Energy-norm strong convergence uniformly for all real time on the mean-zero hierarchy
Physical ADM/Fierz–Pauli convergence	Open source identification	Refinement-compatible physical projector and exact smooth principal-symbol identity absent
Full physical UV completion	Open	Conductance uniqueness, exact smooth principal symbol, full Fréchet Hessian, nonpolynomial fields, composite stress, σ_6 /parallel operator, and smooth renormalization absent

Statement	Status	Exact scope
Device-relative frequency anchor	E2-audited exploratory result	AWS Ramsey, 12 programs and 768 shots; no provider-clock SI traceability
Conditional stress scale	Scoped external-input calibration	CODATA G is input; partial uncertainty only; no gravity measurement
Public provider-metadata traceability	Route-specific NO-GO	Metrology attestation from Rigetti or AWS required
Finite complex carrier	Closed scoped	Actual three-history cycle gives complex dimension 36; ordinary flip also has a nonzero complex sector
Finite source-dagger Born probability	Closed scoped	Normalized squared norm for source-typed projectors in the declared linear cup/cap gluing class
Source $\mathbb{Z}_2$ superselection	Exact derived	Unique nontrivial binary grading class up to complement for the retained generator family
Finite observable dagger algebra	Closed scoped	$M_{18}(\mathbb{C}) \oplus M_{18}(\mathbb{C})$; no cross-charge observable
Finite stochastic quantum engine	Closed scoped	Six-order mixed-unitary channel on canonical compressed controls; raw controls leak from the carrier
Old adjacent-word controlled- C realization	Exact scoped NO-GO	All 720 six-strand Coxeter classes and 108 controller targets exhausted
Marked controlled- C actuator	Minimal typed extension	One generator is necessary and sufficient; positive orientation matching selects its chirality
Exact finite spectral pulse	Exact theoretical derivation	15–32 spectral nodes, degree at most 35; polynomial identity alone is not a circuit
Typed source-word LCU block encoding	Closed scoped extension	Exact 4- or 32-branch source-coefficient LCU for all 32 sector/control cases
Raw-source encoded signal qubit	Exact derived	Rank-36 G_1 eigenspaces at 2 and 3 carry continuous $SU(2) \otimes I_{36}$
Heralded finite actuator	Closed scoped extension	Exact pulse with input-independent success probability approximately $10^{-45.89}$ to $10^{-88.47}$
Deterministic phase-matched amplification	Exact but inefficient	Exact unit success; approximately $10^{23.14}$ to $10^{44.43}$ actuator calls
Low-normalization Chebyshev actuator	Closed scoped extension	Exact all 32 cases; 7–2,793 base calls and 203–39,102 H -block-encoding queries
Source-word Chebyshev walk	Closed scoped extension	Existing LCU unitary, dagger, signal qubit, and success reflection; no explicit square-root defect or spectral read
Phase-frame source-generator compilation	Closed scoped	Exact 34-CNOT, 61-CZ, depth-45 representative with one output S phase
Real-QPU phase-frame execution	Exploratory scoped pass with statistical boundary	Four 256-shot tasks passed frozen sample gates; BASIS population mode not exactly certified

Statement	Status	Exact scope
Matched ordinary QPU control	Exploratory scoped pass	Fresh 998-shot control disfavours projection-only and zero-bias explanation; edge-specific error remains
Full physical engine execution	Open	The 203–39,102-query Chebyshev actuator was not executed; no quantum advantage or gravity measurement
Gravity-apparatus discriminator	Open	No frozen Braid-versus-GR phase, redshift, acceleration, or force residual
Scoped mathematical Braid quantum-gravity theory	Established in declared classes	Internal serial core closed; unrestricted continuum completion and empirical confirmation not claimed
Empirically validated quantum gravity	Not established	Competitor-divergent prediction measurement, external replication, and natural-data confirmation absent
Constructive refutation of OP	Established in-model	Refutes universal priority, not objective reality
Consciousness or empirical ontology claim	Not made	Subjectivity is not phenomenal consciousness

32 Audit chronology and reversals

The derivation was not monotone. Recording reversals is necessary because a final theorem can otherwise conceal which attractive shortcuts failed.

Gate range	Outcome	Scientific role
BGCE261–265	NO-GO / conditional	Stone velocity Hessian, graded trace, orientation, CTP, and raw transfer do not alone derive the Lorentz action.
BGCE266	PASS	Refinement clock fixes $2\pi/3$ and source continuation selects the IR causal tensor.
BGCE267	NO-GO	Fixed-background principal data do not select a unique improved Hilbert stress.
BGCE268–281	mixed to PASS	Quantum channel/carrier intertwiner, finite connection, characteristic-zero tangent, and rank-two local nonmetricity.
BGCE282–284	PASS / partial	Chain rule, chart gluing, and passive local $GL(4)$ tensor naturality.
BGCE285–289	partial	Minimal conservative operator, conductance lift, horizontal uniqueness, $7 + 3$ completion, and fixed-matter rank-seven obstruction.
BGCE290–291	claimed PASS then reversed	The first stress/Ward claim failed uniqueness and nonlinear-integrability red-team checks.
BGCE292–295	PASS	Flat horizontal connection, exponential leaf, typed solder, and finite-to-Hilbert variation.
BGCE296	reversal	Fixed causal grading loses three Lorentz variations.

Gate range	Outcome	Scientific role
BGCE297–299	final scoped PASS	Co-moving grading restores rank ten and removes independent clock coupling; stress/Ward closes in the declared class.
BGCE300R1	full scoped PASS	Noncircular Einstein backreaction and massless spin-2 low-energy sector.
BGCE301R1–303R1	split	$O(h)$ convergence and exact odd cancellation; no global $O(h^2)$ upgrade from Markov analyticity.
BGCE304–305	NO-GO / pivot	Full one-link reflection positivity fails; heat/block routes do not yield finite Hamiltonian evolution.
BGCE306–310	split to PASS	GKSL dilation, native tail, selector gauge removal, and exact temporal total-charge balance.
BGCE311	range NO-GO	Environment-only charges cannot cancel the three tested spatial defects.
BGCE312–324	mixed to PASS	Collision-log dilation, physical-support lift, strong intertwiner, and exact four-component one-collision balance.
BGCE325–336R2	scoped PASS	Effective Hilbert stress, causal CTP influence, continuous event time, and finite reduced balance density.
BGCE337–347	route exclusions	Fixed-conductance, normalizer, and canonical Stinespring-connection routes do not supply the missing nonzero metric variation.
BGCE348	scoped PASS	Gauge-invariant raw/Petz branch contrast equals all four transfer defects and supplies variational interaction stress.
BGCE349–350	scoped PASS	Local 3 + 1 recorded Lorentz parent and exact history-dressed four-component Ward identity.
BGCE351–370	NO-GO / constructive sequence	Excludes parent-alone and homogeneous normalization routes; constructs phase-sensitive control, full-rank record response, finite positive retraction, clock drift, and the joint carrier.
BGCE371	scoped PASS	One parameter-free finite matter–geometry–matter CPTP cycle in the identity-feedback class.
BGCE372–373	PASS / NO-GO	Direct coefficient is 1; natural same-parent maps cannot turn it into the distinct outer-cup ratio 3/5.
BGCE374	NO-GO / conditional PASS	Source-only absolute units are unidentifiable; two external anchors suffice in the declared class.
BGCE375	exact mismatch	Present source is not Standard-Model equivalent.
BGCE376	split PASS	Finite internal Lie algebra is Jacobi-anomaly-free; gravitational first-class constraint closure remains open.

Gate range	Outcome	Scientific role
BGCE377–391	formal closures / route NO-GO	Tests direct internal, projector, incidence, normalizer, and coherent-dilation routes; none yields an identified gravitational first-class algebra.
BGCE392, 441–442	exact route NO-GO	Mean Uhlmann and relative-log forms have zero selected spatial rank; the separable fixed finite algebra has $HH^1 = 0$ and no outer three-plane.
BGCE443	closed scoped	Three Braid-specific closable unbounded local derivations have all-sector rank three, exact $2/3$ constructive gaps, and generated finite-order Lie/Jacobi closure without central anomaly in the derivation representation.
BGCE444	scoped PASS / open	Rank-ten solder induces exact rank-35 quartic capacity with 20-dimensional kernel; actual PBM generation remained open at this gate.
BGCE445	scoped PASS	Compatible finite CPTP channels extend to a strongly continuous AF/quasi-local UCP semigroup with complete bound one.
BGCE446–447	scoped PASS / PASS	The 625-child rule reconstructs fourth jets on the polynomial subspace, and the actual UB612 PBM metric is on the same source-fixed carrier.
BGCE448	scoped PASS / open	Constructs the actual 35-by-55 response and proves midpoint rank 35; the first interval screen did not certify the last mode.
BGCE449	exact PASS	A source-fixed preconditioned minor certifies rank 35 throughout the full saved independent interval box with exact bound below one.
BGCE450–451	NO-GO / partial	Raw/Petz refinement does not supply a physical UV cutoff; a positive source Hodge Laplacian fails child-constant refinement.
BGCE452–454	closed scoped	Hierarchical UCP heat, metric-volume stress/Ward, spectral dimension three, and a device-relative $z = 1$ Lorentzian propagator.
BGCE455–456	split NO-GO to PASS	The raw actual-metric operator fails the Markov sign test; actual-metric positive conductance restores anisotropic refinement-compatible UCP heat and same-operator positive wave evolution.
BGCE457	closed scoped	Every finite operator-logistic identity-feedback iteration remains CPTP, obeys the exact nonautonomous Ward telescope, retains four response directions, and preserves the scoped constraint representation.
BGCE393–438	mixed derivation / NO-GO	Derives anomaly-solution data and representation ingredients while proving that unchanged source operations do not generate the required cube translations.

Gate range	Outcome	Scientific role
BGCE439	final scoped PASS	Derived anomaly-solution groupoid gives three anomaly-free SM character copies, global $\mathbb{Z}_6$, one independent Higgs doublet, and Yukawa interaction types.
BGCE440 R19	E2-audited exploratory PASS	Twelve-program, 768-shot AWS Ramsey packet supplies a device-relative implemented-Braid frequency anchor.
BGCE440 R20	scoped conditional PASS	Official CODATA G populates the second anchor and fixes a conditional stress scale; G is input, not prediction.
BGCE440 R21	route-specific NO-GO	Public AWS metadata lack provider-clock SI traceability and uncertainty; external metrology attestation required.
BGCE458–459	closed scoped	Oriented source history gives a 36-dimensional complex carrier; source reversal and cup/cap counting fix normalized squared-norm probabilities.
BGCE460–464	NO-GO to closed scoped	Naive local generators are reducible; endpoint and Brauer morphisms expose an exact $\mathbb{Z}_2$ charge and close each 18-dimensional sector to its full matrix algebra.
BGCE465–466	split to scoped PASS	Observable-to-control promotion is not automatic; canonical compression nevertheless gives a six-order CPTP/unital stochastic quantum engine.
BGCE467–473	formal/typed construction with retained NO-GO	Reynolds/Zeno compression and a minimal marked controlled- C twirl are finite, but the old adjacent-word algebra cannot realize the control and the twirl does not implement the averaged-Hamiltonian exponential.
BGCE474–477	PASS / route NO-GO	Exact carrier reflection and finite spectral polynomials exist; minimal echo and arbitrary finite-depth old controlled- C actuation fail.
BGCE478–482	typed PASS / obstruction	One block-encoding generator type is sufficient and spectral phases are mathematically exact; old algebraic gates cannot supply the transcendental phases, and source clock flow lacks the required spectrum-transfer morphism.
BGCE483–484	closed scoped extension	Integer source coefficients and one typed source-word select give exact LCU block encodings; the four-probability $1 + 3$ carrier exactly solders to (G_1, H_1, H_2, H_3) .
BGCE485–487	NO-GO to scoped PASS	Minimal scalar QSP and direct compressed V_4 reflections fail; raw G_1 eigenspaces instead supply an exact signal qubit with continuous $SU(2)$.
BGCE488	closed scoped extension	Exact finite heralded Lagrange-LCU actuator for all 32 cases; success probability is approximately $10^{-45.89}$ to $10^{-88.47}$.

Gate range	Outcome	Scientific role
BGCE489	split closed / efficiency NO-GO	Phase-matched amplification gives exact unit success, but black-box query complexity is approximately $10^{23.14}$ to $10^{44.43}$.
BGCE490	closed scoped	Source-compatible Chebyshev conversion improves success by 39.40–87.10 orders and gives exact deterministic circuits with 203–39,102 block-encoding queries.
BGCE491	closed scoped	Hermitianized existing source-word LCU plus success-address reflection realizes the same exact walk without a square-root-defect or spectral-read primitive.
BGCE492–529	compiler sequence	Exact source-action compilation, coordinate compression, phase-cocycle analysis, phase-dressed Yang–Baxter closure, observable covariance, and zero-CNOT decoder retarget reduce the generator to 34 CNOT and 61 CZ gates.
BGCE531–533	exploratory real-QPU PASS / statistical NO-GO	Four 256-shot Cepheus tasks pass frozen sample-level functional gates; exact post hoc audit retains workspace/coherence contrasts but does not certify the BASIS population mode.
BGCE535–547	endpoint and controls	Source-valid endpoint audit, fresh Braid non-certification, matched ordinary-control PASS, calibration/edge audit, and an exact same-skeleton angle-only control isolate but do not eliminate hardware explanations.
BGCE524–552	scoped matter construction	Three-copy generation carrier, source-selected nonnormal response, KMS species strengths, finite odd determinant, nonzero Higgs minimum, relative mass hierarchy, and variational Yukawa bridge.
BGCE556–564	scoped calibration and dressing	Pointed-oriented trine synchronization fixes the dimensionless rate, yields device-relative mass labels and finite source-trace Feshbach poles, while observed labels, RG running, and total uncertainty remain open.
BGCE565–569	conditional neutrino extension / NO-GO	A unital completion supplies three neutral carriers and conditionally fixes $y_\nu = +1$; exact countermodels exclude a dimensionful Majorana scale from the present clock/pair data alone.
BQGEULER-004	closed scoped / physical identification open	Exact refinement intertwining gives two-slot energy-norm strong convergence uniformly for all real time; physical ADM projector and smooth Fierz–Pauli symbol remain underived.

33 Conclusion

The v0.92 result retains the complete v0.9 finite quantum-dynamical, calibration, ultraviolet, constraint, repeated-backreaction, and matter-sector content. The fixed pointed-Braid source supplies not only a Lorentz carrier, matter routing, and the routing ratio $3/5$, but also the infrared causal action selection, a glued continuous metric/nonmetricity tensor, active finite transport, a unique typed physical solder, Hilbert stress, and on-shell Ward conservation in a precise long-wavelength universality class. The source-native gravity and matter variations yield $G_{\mu\nu} = (3/5)T_{\mu\nu}$ in that class, and the flat quadratic sector contains the two-helicity massless spin-2 theory.

The v0.91 foundation retained here lies beneath that gravity limit. The same finite source history reconstructs a 36-dimensional complex carrier, a source dagger and normalized squared-norm projective probability, an exact $\mathbb{Z}_2$ superselection charge, and the observable algebra $M_{18}(\mathbb{C}) \oplus M_{18}(\mathbb{C})$. The construction therefore no longer begins by importing the complete complex quantum formalism. Its boundary is equally explicit: the general dagger-compact mechanism is standard, ordinary flip also generates a complex sector, and empirical identification of this finite carrier with nature remains absent.

The source-selected controls also define a noncommuting finite stochastic engine. A typed source-word extension, exact spectral interpolation, LCU block encoding, and raw-source signal qubit first yield an exact finite heralded actuator. The associated Lagrange black-box amplification is rigorously impractical, requiring approximately $10^{23.14}$ to $10^{44.43}$ calls. BGCE490–491 then move outside that fixed-black-box class: an exact Chebyshev basis change lowers the normalization, and hermitianization of the existing source-word LCU unitary realizes the walk without a square-root-defect or spectral-read primitive. The resulting deterministic circuits use 203–39,102 H -block-encoding queries across all 32 cases.

Version 0.92 takes the first hardware step beyond that theorem. Exact coordinate and phase-frame compilation produces a 34-CNOT, 61-CZ source generator, and BGCE531 executes its frozen probes on Rigetti Cepheus. The sample-level functional packet passes, but exact finite-sample audit does not certify the BASIS population mode. Matched ordinary-control execution disfavours projection-only and simple zero-bias explanations while leaving physical-edge error and general Braid necessity open. This is an exploratory implementation of the finite source generator, not execution of the full Chebyshev actuator, quantum advantage, or gravity evidence.

At finite scale, the source-fixed raw/Petz record contrast supplies the missing interaction-stress variation, a local recorded Lorentzian CTP parent, and an exact four-component history-dressed Ward identity. A full-rank Fisher record action first closes one parameter-free matter–geometry–matter cycle. Its operator-logistic continuation now closes every finite adaptive iteration while preserving CPTP evolution, the exact nonautonomous Ward identity, and four independent global response directions in all eight sectors. The derived anomaly-solution groupoid supplies three anomaly-free Standard-Model chiral character copies, the $\mathbb{Z}_6$ quotient, one independent weak Higgs doublet, and the three charged Yukawa interaction types.

The post-v0.91 matter sequence goes further without fitting observed masses. It selects a nonnormal three-generation response, relative species weights, a variational Yukawa operator, a unique nonzero finite Higgs minimum, and source/device-relative bare and dressed spectral outputs. It also proves nontrivial up/down singular-basis misalignment in a declared class. A conditional unital matter-functor completion supplies exactly three neutral carriers and fixes $y_\nu = +1$ only after unital Frobenius-positive typing. Exact one- and two-clock-unit countermodels show that the present clock and normalized Majorana pair do not determine a dimensionful Majorana action scale.

The ultraviolet advance is likewise exact but scoped. The compatible source-cylinder channels extend to a norm-one AF/quasi-local quantum semigroup at every refinement depth. The rank-ten metric solder, source-native 625-child fourth-jet rule, fixed actual-PBM carrier intertwiner, and rational interval certificate give a full-rank actual UB612 quartic response across its saved coefficient box. The actual inverse spatial metric now also generates a strictly positive directional-conductance hierarchy whose same positive operator supplies refinement-compatible anisotropic UCP heat, positive wave evolution, and spatial spectral dimension

three. This closes the registered all-scale, quartic-response, and anisotropic-UCP packages; it does not close arbitrary nonlinear composite observables or general physical smooth-spacetime UV completion.

BQGEULER-004 adds exact trajectory control on this hierarchy. The two-slot mean-zero wave groups converge strongly in the energy norm uniformly for all real time, with error exactly equal to the initial energy tail. What remains is not another finite-time estimate but physical identification: no source-derived refinement pullback yet identifies those two slots with the ADM quotient, and the exact smooth Fierz–Pauli principal symbol is still open.

The constraint result closes a second previously central internal gate. Three source-generated local derivations survive refinement as independent closable unbounded generators, differ from the unsigned reference in every signed-sector direction, and generate an exact finite-order Lie algebra with vanishing Jacobi residual and no central anomaly in the derivation representation. This is an anomaly-free source-generated constraint algebra in its declared scope. It is not yet an identification with ADM Hamiltonian and momentum constraints or the hypersurface-deformation algebra.

The new calibration result is operational. An independently E2-audited AWS Ramsey execution supplies a nonzero device-relative implemented-Braid frequency anchor and registered time, length, and energy units. Official CODATA G then conditionally fixes the stress unit under the frozen two-anchor map. This removes the scale degeneracy only after external inputs are supplied. It neither predicts G nor measures a Braid gravitational response. Public AWS metadata do not expose enough clock metrology to claim a complete SI uncertainty budget.

The result is not promoted beyond its proof. The finite first-variation residual is only bounded by $O(h)$. The parent is not full ten-component nonlinear gravity; the feedback and conductance laws are not proved unique; and no uniform nonlinear infinite-iteration theorem is claimed. The UV theorem does not yet cover the full metric Fréchet Hessian, arbitrary nonpolynomial sampling, bounded composite stress, the complete σ_6 /parallel operator, an exact smooth principal symbol, or smooth-spacetime renormalization. The direct finite coefficient 1 and full-corner routing ratio 3/5 are exact but differently typed. The matter construction does not yet supply observed particle labels, a universal natural mass scale, numerical CKM/PMNS values, RG flow, total uncertainty, or empirical agreement. The finite actuator requires a typed source-word controller and success-address mark, and the full circuit has not been executed. Constraint moment-map/ADM typing, full calibration, a Braid-specific gravimeter or clock residual, and empirical gravity remain untested.

The philosophical conclusion is constructive. Complex quantum kinematics, projective probability, objective Lorentz structure, conserved stress, and a classical gravity limit need not all be primitive in every exact model; they can be generated in stages from nonidentical, marked, standpoint-bearing processes and their lawful intersection. Objectivity is retained as stable public invariance. What is rejected is its universal ontological priority. The physical conclusion is correspondingly measured: this is a mathematically closed Braid quantum-gravity theory candidate in its declared finite and source-cylinder scopes, not yet an unrestricted continuum completion or empirically validated theory.

Retained v0.7 derivation detail

The following appendices retain the still-operative finite-source, locality, metric-affine, action, tensor-carrier, regeneration, comparison, falsification, and objection analyses on which the v0.92 advance depends. Superseded v0.7 bridge status, the old conditional MMR2/SDPC route, and routes later reversed or replaced are not reproduced here. Where a retained appendix describes the finite parent, four-current balance, matter sector, constraints, calibration, ultraviolet carrier, or repeated backreaction as open, the v0.92 main text above supersedes that historical boundary only to the exact scope stated here. The v0.92 main text is the sole current claim ledger.

A A source-first pointed-braid setting

A.1 Finite source, nonterminal law

Let $\mathcal{A}_n = M_{5^n}(\mathbb{C})$ denote the finite matrix algebra at stage n , with the refinement

$$j_n : \mathcal{A}_n \longrightarrow \mathcal{A}_{n+1}, \quad j_n(X) = X \otimes I_5. \quad (150)$$

Every stage is finite. The sequence has no terminal stage. This distinction is important: a nonterminal tower is not, by itself, a continuum, an infinite-mode quantum field, or physical spacetime. It supplies an exact setting in which one law can be tested at arbitrarily many finite stages.

The source is pointed and braided: it carries a distinguished state and typed unitary braid transports satisfying the declared Yang–Baxter, insertion, and composition laws. The all-orders endpoint-consistency theorem and the positive bilateral response operators were established previously [16]. The present paper uses that source theorem but does not infer physical subjectivity from it.

The computational realization contains eight signed source sectors. Each sector supplies the primitive source packet

$$Q_2 = (R, H, F, P), \quad (151)$$

where the symbols denote the registered relation and source fields, not independently fitted target geometry. Three selected signed-braid generators act as commuting involutions on the certified orbit. From one four-state source slice they generate the opposite slice. Exact checks recover all 12 declared source updates, signed relations, clock fields, and associated source covectors; the identity-map comparator fails in all 12 cases.

A.2 Source-native geometry before the present carrier

The relevant prior source chain is

$$(R, H, F, P) \longrightarrow \text{signed-braid source orbit} \longrightarrow \text{affine split-frame geometry} \longrightarrow \text{coframe and holonomy}. \quad (152)$$

The end-to-end constructor uses no stored future geometry. It reconstructs the connection data, split-frame defect, current term, affine edge, future spatial edges, and future faces from the predicted source relation. In the certified fixed orbit it recovers 12/12 source-slice updates, 24/24 affine edges, 24/24 future spatial edges, and 6/6 future faces exactly. Removing either of two required source sectors breaks the corresponding controls.

This result removes one important circularity: the geometry used below is not inserted as a target answer and then redescribed as braid-generated. It is the output of the registered finite source constructor. The constructor remains global on the fixed finite orbit rather than a strictly local Cauchy law. Version 0.5 separately removes the numerical ambient-dimension input at the carrier layer, as follows.

A.3 A target-free 1 + 3 carrier and its affine completion

Let the source-derived observer quotient be $Q_{\text{obs}} = V_4$, with intrinsic affine action

$$\text{AGL}(2, 2) \cong S_4.$$

Start over the rational source layer and let $W_{\mathbb{Q}} = \mathbb{Q}^{V_4}$ be the space of observer readouts. Impose three explicit admissibility conditions: the readout contains the constant order unit, separates all four quotient states, and is invariant under the full intrinsic affine action. The S_4 permutation commutant is spanned by I and the all-ones matrix J . Its equivariant idempotents have ranks 0, 1, 3, 4. Unitality excludes ranks 0 and 3,

while point separation excludes rank 1. Therefore the minimal admissible readout is the full carrier, without supplying a target dimension:

$$W_{\mathbb{Q}} = \mathbb{Q}\mathbf{1} \oplus W_0, \quad \dim(\mathbb{Q}\mathbf{1}, W_0) = (1, 3). \quad (153)$$

The three-space W_0 is the irreducible mean-zero representation. All 24 affine observer actions preserve this split.

The finite Braid data do not by themselves contain the real numbers as a completed scalar field. We therefore state one completion principle explicitly.

Assumption A.1 (Observer Uniform Cauchy Closure). Observer readouts and the marked clock direction are closed under uniform Cauchy limits without adding independent observer directions.

Under Assumption A.1, scalar completion gives

$$W = \mathbb{R}^{V_4} = \mathbb{R}\mathbf{1} \oplus W_{0,\mathbb{R}}, \quad \dim W = 4, \quad (154)$$

with the same 1+3 split. Every finite-dimensional real vector space canonically determines an affine smooth carrier $\text{Aff}(W)$: choosing an origin gives a global chart, changing that origin is only a translation, and

$$T_x \text{Aff}(W) \cong W \quad (155)$$

for every x . Thus neither the numeral four nor a separate ambient four-manifold is a primitive mathematical input. A declared cup/null composition additionally gives the dimensionless Lorentz form

$$g_D = \frac{16}{75} \text{diag}(-3, 1, 1, 1), \quad (156)$$

whose four evaluation rays are null. Physical rods, seconds, nonconstant curvature, and the identification of points of $\text{Aff}(W)$ with physical events do not follow from this algebraic completion.

The source path and the carrier have different roles. The path indexes inclusions and disjoint supports in the operator net; it is not renamed as three spatial axes. The three relative directions in Eq. (153) are internal observer-readout directions. Later finite-event audits establish a distinct formal event identification inside the fixed mathematical model without claiming calibration to physical spacetime.

A.4 The source also closes a finite gravity equation

The source-native construction does not stop at a static geometry. On each certified edge (m, d) it obeys the exact first-order coframe equation

$$E_{m \oplus d} = C_{m,d} E_m R_d + K_{m,d}[C, j, F], \quad (157)$$

with the second covariant face closure

$$D_C K = F_C E. \quad (158)$$

Here $C_{m,d}$ is the source-generated ECDR/BSFG connection, R_d reverses the crossed generator orientation, j is the IBC current, F is the affine face curvature recomputed from (C, j) , and $K[C, j, F]$ is the coframe source that cannot be produced by invertible transport alone.

On the fixed eight-source orbit, Eq. (157) recovers 24/24 future coframes, 24/24 nonzero source terms, and all future ranks exactly; Eq. (158) holds on 12/12 faces and is nonzero on 10/12. Transport without K fails on all 12 rank-changing edges, while zeroing the curvature fails on all 40 nonzero-curvature columns. The source term is therefore not a removable decoration of a stored geometry.

For the induced spatial Gram matrix $G_m = E_m^\top g_m E_m$, the same update yields the finite metric-affine equation

$$G_n = R_e^\top G_m R_e + \mathcal{N}_{e,E} + \mathcal{L}_{e,K} + \mathcal{Q}_{e,K} \quad (159)$$

with

$$\mathcal{N}_{e,E} = R_e^\top E_m^\top (C_e^\top g_n C_e - g_m) E_m R_e, \quad (160)$$

$$\mathcal{L}_{e,K} = (C_e E_m R_e)^\top g_n K_e + K_e^\top g_n (C_e E_m R_e), \quad (161)$$

$$\mathcal{Q}_{e,K} = K_e^\top g_n K_e. \quad (162)$$

All three terms are nonzero and separately necessary on 24/24 edges. Thus the connection's nonmetricity, the source-linear response, and the finite source-quadratic response cannot be collapsed into one another. This is a source-generated finite gravitational dynamics in the declared model. Its packet lifts by one fixed law to every finite braid stage, but only as a stage-compatible cylinder section: it does not yet generate new local spacetime cells.

Equations (157)–(159) are not themselves the Einstein equation $G_{\mu\nu} = \kappa T_{\mu\nu}$. Calling them finite Braid gravity equations records their exact source–geometry dynamics while preserving that distinction. The next construction gives a direct but explicitly conditional local connection to the Einstein equation.

B From finite updates to source-generated locality

B.1 A finite source-graph local net

At stage n , let G_n be the graph whose vertices are tensor factors and whose edges join adjacent noncommuting Braid generators. In every certified signed sector, distant generators commute exactly and adjacent triples obey the Yang–Baxter relation. For a vertex set U , define

$$\mathfrak{A}_n(U) = \text{End}((\mathbb{C}^5)^{\otimes U}) \otimes I_{U^c}. \quad (163)$$

Then $U \subseteq V$ implies $\mathfrak{A}_n(U) \subseteq \mathfrak{A}_n(V)$, disjoint supports commute, and a depth- t adjacent-Braid update expands support by at most graph distance t . The refinement j_n is strict and unital. These are exact finite isotony, locality, and finite-propagation statements on a source-generated graph. They are not yet Haag–Kastler locality on a Lorentzian manifold, and graph distance is not identified with physical distance.

The positive Braid generator candidate

$$L_n^B = \sum_i (I - \text{Ad } R_i) \quad (164)$$

has a refinement-boundary defect

$$\Delta_n(X) = L_{n+1}^B j_n(X) - j_n L_n^B(X). \quad (165)$$

Its normalized boundary reduction is nonzero in all eight sectors, has 41 nonzero matrix entries and squared Frobenius norm 108/5, fills all six registered boundary-response channels, and returns to zero after the inverse crossing. Thus the stable carrier and the local generation event are distinct outputs of the same source lineage.

B.2 A finite Ward law before physical stress-energy

The boundary response, the existing IBC edge data, and the ordered-edge memory admit a six-component cell variable. If d_n is the incidence matrix of the stage path, $D_n = d_n \otimes I_6$, and H is the exact rank-six cell Gram matrix, define

$$\mathcal{S}_n(q) = \frac{1}{2} (D_n q)^\top (I_{n-1} \otimes H) (D_n q), \quad L_n = D_n^\top (I_{n-1} \otimes H) D_n. \quad (166)$$

For every $n \geq 2$,

$$\text{rank } L_n = 6(n-1), \quad \ker L_n = \mathbf{1}_n \otimes \mathbb{F}^6, \quad (\mathbf{1}_n^\top \otimes I_6)L_n = 0. \quad (167)$$

The last identity is an exact all-finite-stage constant-shift Ward law. For every graph interval, interior edge fluxes cancel and only the two cut-boundary fluxes remain. The incidence mechanism is not uniquely Braid-specific; the Braid-specific content lies in the rank-six cell packet and its actual-versus-flip response. Equation (167) is weaker than stress-energy conservation and does not constitute a spacetime Bianchi identity.

C Continuous cells and the native metric-affine sector

C.1 Stable-square gluing, metric faces, and curvature

The finite edge transports can be glued on stable squares with exact agreement on overlaps. On the source-native cubical cells, the generated Gram data admit continuous face representatives, and their compatible edge transports define a continuous connection with nonzero rank-four curvature on the certified face packet. A common projective timelike line and a $1 \oplus 3$ splitting can then be selected consistently on the regular sector. The induced spatial connection retains nonzero curvature.

These statements produce continuous representatives on certified finite cells; they do not derive a global smooth manifold, an arbitrary-field connection, or a physical continuum limit. A separate refinement analysis also shows why this distinction matters: the current tensor-tower embedding preserves packets but does not subdivide geometric edges, while the exact affine edge refinement is endpoint-blind to the nonuniform off-shell defects needed to select an action by refinement alone.

C.2 Torsion and nonmetricity are not removable errors

The full source connection splits into a metric-compatible polar core and a metric-symmetric part. Both curvature components scale at the same leading area order on the presently covered regular faces. Neither the tested perturbative contortion correction nor the exact finite single-face torsion-free ansatz closes. The unmodified construction therefore does not converge to the Levi-Civita connection.

This negative result selects a wider native geometry rather than eliminating gravity. Write the regular-face source coordinate as

$$\hat{K} = I - A, \quad A = UP. \quad (168)$$

With $\epsilon = \text{sign det } U$, $R = \epsilon U$, $\rho = (\det P)^{1/3}$, and $S = \rho^{-1}P$, one has the unique reconstruction

$$A = \epsilon R \rho S, \text{quad } \det S = 1. \quad (169)$$

The connected sectors have dimensions

$$\underbrace{3}_{\text{rotation}} + \underbrace{1}_{\text{dilation}} + \underbrace{5}_{\text{unimodular shear}}, \quad (170)$$

together with the discrete parity ϵ . All three continuous sectors are nontrivial in all 16 regular-face records, and their product reconstructs A exactly. This is precisely the representation type expected for a post-Riemannian affine source. The names “spin,” “dilation current,” and “shear hypermomentum” are not physical identifications here; no matter current has yet been measured or derived.

C.3 Stratified junctions

Rank-changing paths do not destroy the entire tangential metric. At the four exceptional junction records, after removal of the declared endpoint double zero, the determinant is exactly a nonzero constant times $(t - r)^2$ over $\mathbb{Q}(\sqrt{17})$, with $0 < r < 1$ in both real embeddings. At r the full Gram matrix has rank two and inertia $(0, 2, 1)$: two positive directions and one zero direction, with no negative direction. The metric is positive definite on both adjacent open intervals. The isolated event is therefore a codimension-one channel loss with a positive two-dimensional quotient, not a collapse of the tangential sector. Interpreting it as physical dimension creation would require an additional bridge and is not claimed.

D A source-native finite first-order action

D.1 Why a metric-only action is insufficient

The full Braid boundary load decomposes into zero clock response and nonzero rotation, dilation, and shear responses. A functional depending on the spatial coframe E only through the Gram matrix $E^\top g E$ has variation in the six-dimensional image

$$\{gES : S^\top = S\}. \quad (171)$$

The rotation response lies instead in the independent three-dimensional image

$$\{gER : R^\top = -R\}. \quad (172)$$

The two images intersect only at zero, while the certified full load has nonzero components in both. Therefore the source response cannot be generated by a metric-only action. Coframe and connection variables must remain independent at the finite level.

D.2 The full-trace action

For each oriented face f , let $\mathcal{R}_f$ denote the history-dressed linear curvature obtained from the two ordered boundary paths, let E_f denote the corresponding source-generated base coframe, let $E_\top$ be the common-top coframe, and let $g_\top$ be the generated top metric. The source-native full-trace action is

$$\mathcal{S}_B[E, C, g] = \sum_{f \subset \partial e} \sigma_f \operatorname{Tr}(E_\top^\top g_\top \mathcal{R}_f[C] E_f), \quad (173)$$

where σ_f is the orientation fixed by the cell boundary. Equal column weights are forced, up to one overall scale, by covariance under the generating permutations of the three spatial columns. No face weight, sector weight, prefix-dependent scale, X51 residual, or Einstein tensor is used.

On the certified packets, the action is nonzero for both prefixes. Its analytic coframe variation agrees with exact central differences in 192/192 components, and its analytic connection variation agrees in 384/384 components. The top coframe Euler response reproduces the independently frozen full boundary solder for both prefixes, including the rotation, dilation, and shear sectors. The connection Euler response is nonzero on all 12 active edges for each prefix. Thus one finite action produces both coframe and connection equations.

The action is not a fitted scalarization of the affine translation block. Affine-origin invariance eliminates the translation scalar and selects the linear coframe-curvature pairing up to normalization. Earlier direct attempts to identify a face scalar, a Cartan three-cell scalar, or a full curvature solder with the prior X51 residual all failed by exact non-collinearity. Those no-go results are retained: Eq. (173) is justified by its own source covariance and variation, not by reverse-engineering X51.

D.3 Exact local-frame Noether identity

Let $\mathcal{E}_E$, $\mathcal{E}_C$, and $\mathcal{E}_g$ denote the coframe, connection, and top-metric Euler responses of Eq. (173). Under a local infinitesimal frame change at a vertex, the incident coframes and connections transform together. Invariance of the action gives the matrix identity

$$\mathcal{N}_v(\mathcal{E}_E, \mathcal{E}_C, \mathcal{E}_g) = 0 \quad (174)$$

at every vertex, where $\mathcal{N}_v$ is the signed sum of the coframe contribution, incident incoming and outgoing connection contributions, and, at the top vertex, the metric contribution. The identity holds exactly at all 14 non-top vertex tests and both top-vertex tests. With $g_\top$ fixed, the full $GL(4)$ identity fails as it should, while all 12 tested generators preserving $g_\top$ retain the Lorentz-type subidentity. Finite local-frame covariance also passes 16/16 tests.

Equation (174) removes the previous absence of an action-level conservation structure. It does not make the Euler responses vanish. A Noether identity relates field equations; it is not itself a solved field equation or a physical energy current.

D.4 A covariant-defect variational connector

The existing IBC, ECDR, BODC, and history-curvature library was tested as a possible independent balance action. A fixed opposite source cancels the action value, top-coframe response, and metric response at the evaluated point, but leaves eight base-coframe masks and 24 connection-edge responses. Recomputing the opposite term dynamically cancels everything only because it makes the total action identically zero. That is a tautological negative copy, not an independent physical source.

The resulting no-go did not establish that the Braid source lacks a variational connector. On an oriented edge $e = (m, d)$ with high vertex $n = m \oplus d$, define the independently generated covariant coframe defect

$$K_e = E_n - C_e E_m R_d. \quad (175)$$

This is the same X36 defect reconstructed from the source on all 24 certified edges. Its coefficient-one quadratic action is

$$\mathcal{S}_D[E, C, g] = \frac{1}{2} \sum_e \text{Tr}(K_e^\top g_n K_e). \quad (176)$$

It was frozen without reading the curvature-action Euler residual, X51, or any Einstein tensor.

The connector is nontrivial and off shell. Its exact analytic derivatives agree with central differences in 832/832 tested entries across coframe, connection, and metric slots. The induced map has rank 12 on every full defect space and rank 8 on every traversed-column-zero source subspace; the actual cotangent image is nonzero on 24/24 edges. Finite local-frame covariance holds in 16/16 tests, the infinitesimal vertex Noether identities hold in 256/256 tests, and both off-shell perturbations change the action. Thus Eq. (176) supplies the previously missing source-to-cotangent typing without fitting a negative copy.

The current result is therefore

$$\boxed{\text{Braid source} \rightarrow \text{geometry} \rightarrow (\mathcal{S}_B, \mathcal{S}_D) \rightarrow T_{(E, C, g)}^*, \quad \delta(\mathcal{S}_B + \mathcal{S}_D) = 0 \text{ untested.}} \quad (177)$$

This retained finite-action calculation supplies the source-to-cotangent typing used by the later v0.8 analysis. Its historical coefficient-one stationarity question is no longer presented as the current completion route: the main text replaces that route with source-selected continuum action, typed solder, Hilbert stress, and Ward results in the declared infrared class. Exact full-finite stationarity remains a distinct ultraviolet question.

E Why scalar invariants are not enough

E.1 Soldered source holonomy

For each prefix $p \in \{A, B\}$, let

$$E_p \in \text{Mat}_{4 \times 3}(\mathbb{R}), \quad \text{rank } E_p = 3,$$

be the source-generated coframe, let $g_p \in \text{Mat}_{4 \times 4}(\mathbb{R})$ be the ambient symmetric form, and let $H_p \in GL_3$ be the internal face holonomy. Define the Gram matrix

$$M_p = E_p^T g_p E_p, \quad (178)$$

assumed invertible on the certified regular stratum.

The source-only update has two independently meaningful descriptions. The internal path mismatch is H_p , while the ambient coframe curvature is the difference between the two source-generated paths to the same face. Exact certificates show, in every one of the eight face packets, that both paths reach the same target coframe while the internal holonomy is nonidentity, and that the internal action, ambient curvature action, and source curl agree. A zero-source control fails in all eight packets.

This is not yet Riemann curvature or Einstein gravity. It is a finite source-only holonomy/curvature identity on a regular stratum. Its role here is to provide directional data that is both nontrivial and traceable to the braid source.

E.2 The scalar route and its exact obstruction

Conjugacy-invariant characteristic data of H_A and H_B provide three scalar channels per prefix. The product of their characteristic polynomials gives six nonleading coefficients that are invariant under exchange $A \leftrightarrow B$ and distinguish both one-sided collapses in the certified packets. This is a valid relational readout, but it discards direction.

The loss is not merely aesthetic. For the declared signed-permutation frame group acting on Sym_3 , the fixed subspace is one-dimensional, spanned by the identity. Therefore any full-frame-equivariant linear map from six trivially transforming scalar channels into one anisotropic Sym_3 block has rank at most one. The exact intertwiner calculation has constraint rank 30, a six-dimensional map space, and maximum anchor rank one. Moreover, the original modular residues have no unique real lift and carry no stage index.

Proposition E.1 (Scalar-anchor no-go). *Within the declared linear, signed-permutation-equivariant anchor class, six presentation-invariant scalar channels cannot determine a full-rank anisotropic symmetric three-tensor. A covariant directional carrier must be formed before the directional information is quotiented away.*

The proposition is scoped. It does not forbid nonlinear maps, separately typed scalar observables, or enriched covariant carriers. It instead identifies the exact repair used next.

F The generated tensor carrier on the source-derived 1 + 3 type

F.1 Construction

The formulas below were originally written in a declared $\mathbb{R}^4$ basis. In v0.5 their coordinate-free type is the source-derived carrier W of Eq. (154): $E_p \in \text{Hom}(\mathbb{R}^3, W)$, $g_p \in \text{Sym}^2(W^*)$, and $T_C \in \text{Sym}^2(W)$. A chosen constant-plus-character basis identifies these spaces with the displayed 4×3 and 4×4 matrices. Thus the component calculations are unchanged while the numeral four is no longer a primitive target input.

Retain the holonomy defect until it has been soldered to the ambient coframe:

$$D_p = E_p(I - H_p). \quad (179)$$

Define

$$T_p = D_p M_p^{-1} E_p^\top + E_p M_p^{-1} D_p^\top \in \text{Sym}_4(\mathbb{R}), \quad (180)$$

and define the joint or intersection carrier

$$T_C = T_A + T_B. \quad (181)$$

The construction performs the quotient in the correct order. The inverse Gram matrix removes internal-basis dependence, while the coframe preserves ambient direction. Symmetrization yields a rank-two ambient carrier. Only then are the two standpoint contributions joined.

F.2 Transformation laws

Let $G_p \in GL_3$ be independent internal changes of basis. Under

$$E_p \mapsto E_p G_p, \quad H_p \mapsto G_p^{-1} H_p G_p, \quad (182)$$

we have $D_p \mapsto D_p G_p$ and $M_p \mapsto G_p^\top M_p G_p$. Substitution in Eq. (180) gives

$$T_p \mapsto T_p. \quad (183)$$

Thus the carrier does not depend on either prefix's internal presentation.

Let $L \in GL_4$ be an ambient basis change, with

$$E_p \mapsto L E_p, \quad g_p \mapsto L^{-\top} g_p L^{-1}. \quad (184)$$

Then M_p is unchanged and

$$T_C \mapsto L T_C L^\top. \quad (185)$$

This is covariance, not componentwise invariance: the object is stable while its components change tensorially.

Finally, exchanging A and B leaves Eq. (181) unchanged. Crucially, exchange invariance is not identification of the two inputs. The certified prefix carriers are distinct, and replacing the pair by (A, A) , (B, B) , or identity holonomy changes the result.

Proposition F.1 (Structural carrier properties). *Whenever M_A and M_B are invertible, T_C is symmetric, invariant under independent internal GL_3 presentation changes, invariant under prefix exchange, and covariant by ambient congruence.*

Proof. Symmetry follows directly from Eq. (180). The internal transformation rules imply

$$D_p G_p (G_p^{-1} M_p^{-1} G_p^{-\top}) G_p^\top E_p^\top = D_p M_p^{-1} E_p^\top,$$

and similarly for the transpose term. Exchange invariance follows from commutativity of addition. Ambient covariance follows by substituting $L E_p$ and $L D_p$ into Eq. (180). $\square$

F.3 Exact finite certificate

Theorem F.2 (Certified joint-carrier theorem). *For the registered actual source packets and their source-generated geometry:*

1. T_A and T_B are nonzero symmetric rank-three carriers;
2. T_C is nonzero, symmetric, and rank three;
3. all ten native components of $T_C \in \text{Sym}_4$ are nonzero;
4. T_C is exactly invariant under independently selected internal GL_3 gauges and under A/B exchange;
5. the congruence law in Eq. (185) holds exactly;
6. T_C differs from $2T_A$, $2T_B$, and the identity-holonomy zero control;
7. all irrational $\sqrt{17}$ parts cancel, so T_A, T_B, T_C descend exactly to rational matrices.

The theorem is an exact fixed-finite result, not a fitted numerical approximation. Items 4–6 jointly matter for the interpretation. Presentation redundancy disappears, ambient direction remains, and neither source standpoint is dispensable.

The ten components have the same mathematical type as an existing ten-component metric-response interface in the larger unification program. Equality to that response, physical normalization, units, insertion into the 87-component state, and laboratory implementation have not been proved. Accordingly, T_C is called a *carrier*, not an observed metric or stress tensor.

G Regeneration at every finite stage

G.1 Observation mates

Let $\phi : M_5(\mathbb{C}) \rightarrow \mathbb{C}$ be any state satisfying $\phi(I_5) = 1$, and define

$$E_\phi = \text{id} \otimes \phi. \quad (186)$$

Then

$$E_\phi \circ j_n = \text{id}_{\mathcal{A}_n}. \quad (187)$$

No tracial assumption is required on the refined range. The normalized trace, a point state, and a nontracial full-support state therefore recover the same embedded source record.

For an observer word w represented by a unitary U_w , define the co-moving refinement and readback

$$J_{w,n} = \text{Ad}(U_w) \circ j_n, \quad E_{w,\phi} = E_\phi \circ \text{Ad}(U_w^{-1}). \quad (188)$$

Then $E_{w,\phi} J_{w,n} = \text{id}$.

Let $\mathcal{C}_n$ denote the deterministic composite that maps the source packet at stage n through the source-orbit constructor, geometry, holonomy, and Eq. (181) to its ten-component carrier.

Theorem G.1 (All-finite-stage carrier naturality). *For every finite stage n , every declared observer word w , and every unital spectator-factor state ϕ ,*

$$\mathcal{C}_{n+1} \circ J_{w,n} = \mathcal{C}_n \quad (189)$$

after the co-moving readback of Eq. (188). The same carrier law and the same coefficients are used at every stage; no stage-dependent refit is permitted or required.

Proof. Equation (187) recovers each refined source component exactly. Conjugating the refinement and composing with the inverse conjugation preserves that identity. The source-orbit, geometry, holonomy, and

carrier maps are deterministic functionals of the recovered source packet, hence their composition recovers the same carrier. The exact regression checks cover eight actual source sectors, four source components, and three distinct observation states: 96/96 depth-one readbacks and 96/96 boundary co-moving re-entries. Representative depth-one and depth-two checks pass 6/6 with the same ten-channel carrier hash. $\square$

The theorem is stronger than preservation under a chosen normalized trace: the carrier is independent of which unital spectator state is used on the refined range. It is also weaker than a physical continuum theorem. No limit topology, infinite-mode Hilbert space, or continuum field is inferred from Eq. (189).

H What is created, and what is not

H.1 Objectivity survives, but changes status

The result does not replace objectivity with private experience. The output is public in the relevant mathematical sense: internal presentation choices do not alter it, observer-word refinements admit exact co-moving recovery, and ambient changes act covariantly. What changes is the ontological order. The public structure is the result of a quotient-and-soldering operation, not the assumed stage on which that operation occurs.

This distinguishes generated objectivity from a naive averaging of perspectives. The carrier is not the common scalar part left after every direction is removed. Proposition E.1 shows that this route is too poor to recover an anisotropic tensor. The construction must retain the directional defect of each standpoint, solder it to a common ambient type, and only then quotient the internal descriptions. Objectivity is achieved by a disciplined relation among standpoints, not by erasing them.

H.2 The role of A , B , and C

Formally, A and B are inequivalent source prefixes with distinct certified holonomies and carriers. Their exchange is a symmetry of the joint output, but they are not identified. C is the joint carrier after internal presentation dependence has been removed. In the proposed ontology, this formal pattern is read as follows:

- A is Absolute Subjectivity, associated with light;
- B is Relative Subjectivity, an impersonal mode associated with time or a clock, not an individual, organism, mind, or conscious observer;
- C is Intersection Subjectivity, associated with space and objective manifestation.

The formal theorem establishes the grammar “distinct A and B generate a nontrivial invariant/covariant C .” It does not establish those physical nouns. Within the ontology, however, B is explicitly pre-personal and pre-biological: relative clocking and ordering are held to belong to the universe before humanity. This is the precise boundary between mathematical formulation and ontological interpretation.

H.3 Creation as regeneration rather than first manufacture

The all-stage law changes the meaning of creation. It is not a single event in which a permanent object is made once and then carried unchanged forever. At each finite stage, a refined source packet can be read back co-movingly and the same law regenerates the carrier. Thus the proposal is compatible with a world whose apparent persistence is repeated structural re-creation.

This does not mean that every stage produces a newly measured nonzero excursion. The fixed-window boundary response is nonzero in all eight certified sectors, but a nonzero boundary excursion of T_C itself has not yet been proved. The theorem establishes exact regeneration of the carrier law, not a complete physical dynamics of creation events.

I Relation to nearby programs

Program	Shared concern	Difference from the present claim
Invariantism and structural realism	Objective content is associated with invariants or relational structure.	The present model asks whether the invariant must be primitive. Its invariant/covariant carrier is constructed only after distinct source standpoints contribute directional data.
Relational quantum mechanics	Physical facts or states are relative to physical systems rather than globally absolute.	The present result is not an interpretation of the quantum state. It constructs a specific real Sym_4 carrier and an all-stage regeneration law from a pointed-braid source.
Quantum reference frames	Descriptions transform between physical frames without an external absolute frame.	QRF work supplies precise perspective transformations within quantum kinematics. Here the primary claim concerns the generation of a shared tensor carrier from source intersection, not merely switching among descriptions of a pre-specified system.
QBism	The quantum state is not an objective property; interaction and agent-relative probabilities are central.	“Subjectivity” here is not belief, experience, or conscious agency. The output is an exact inter-presentation tensor, not a personal probability assignment.
Perspectival realism	Reliable knowledge can arise across a plurality of situated perspectives.	Perspectival realism is principally an account of scientific representation and inquiry. The present paper proposes a formal generative ontology and a source-level physical research program.
Braided particle models	Braid topology can encode particle-like structure and interactions [18].	The present use of braiding concerns coherence, source transport, holonomy, and generated objectivity; it does not identify Standard-Model particles with braid words.
Set-theoretic Yang–Baxter models	Permutations of $X \times X$ can satisfy Yang–Baxter independently of this model [19].	The 25-event permutation is not claimed as a new class of Yang–Baxter solution; its fixed spectral-marker provenance and distinction from identity/flip are the checked features. No classification-level inequivalence is established.
Quantum causal histories	Locally finite pre-spacetime causal relations can carry matrix algebras and completely positive evolution [21].	The present source begins with a pointed-braid matrix-algebra tower and tests a particular spectral event action. It has not derived a physical causal continuum or shown an Einstein limit from finite evolution.
Causal-set growth	Discrete causal order and covariance can support sequential growth dynamics [26].	The present address cylinders and braid crossings have different primitives; neither a causal-set embedding nor a physical growth law is proved.
Thermodynamic Einstein route	The Einstein equation can follow from local horizon thermodynamic assumptions [23].	The present route instead assumes a Palatini action and matter/Ward conditions on a selected smooth domain; the source-to-domain convergence is still open.
Metric-affine gauge gravity	Coframe, independent connection, torsion, nonmetricity, hypermomentum, and Noether identities form a first-order gravitational framework [6].	The present work does not assume a continuum affine gauge theory as its microscopic source. It gives finite affine-source structures and actions, but the actual finite-source-to-smooth limit is not yet verified.

Program	Shared concern	Difference from the present claim
Earlier Subjectivity Intersection work	Pointed braids provide all-orders intersection consistency and compositional objectivity [16].	Version 0.8 retains the finite geometry and action chain and adds source-derived 3/5, an exact Lorentz carrier, continuous tensor transport, stress/Ward, and the scoped Einstein/spin-2 limit.

The novelty claim is therefore conjunctive. Neither relationality, perspective dependence, invariance, braid groups, nor process-like ontology is individually new. The contribution is the exact chain

$$\begin{aligned}
& \text{distinct braid standpoints} \longrightarrow \text{target-free } 1 + 3 \text{ carrier} \longrightarrow \text{source-only holonomy} \\
& \longrightarrow \text{direction-preserving soldering} \longrightarrow \text{joint covariant carrier} \\
& \longrightarrow \text{finite metric-affine dynamics} \longrightarrow \text{curvature and defect actions} \\
& \longrightarrow \text{finite spectral events and a selected smooth-domain matter action} \\
& \longrightarrow \text{source-selected infrared action, stress, and Ward identity} \\
& \longrightarrow \text{scoped Einstein and massless spin-2 limit,}
\end{aligned} \tag{190}$$

together with an explicit philosophical argument showing what this chain does and does not refute.

J Operational content and falsification

J.1 A reproducible mathematical readout

The current operationalization is a deterministic computational readout, not a laboratory apparatus. Given the registered source packet, it performs the following steps:

1. generate the opposite source slice using the frozen signed-braid action;
2. reconstruct the affine geometry and the two source-prefix coframes;
3. compute the source-only face holonomies;
4. form D_p , M_p , T_p , and T_C by Eqs. (178)–(181);
5. output the ten independent entries of $T_C \in \text{Sym}_4$;
6. verify internal-gauge invariance, prefix-exchange invariance, ambient congruence, and the three collapse controls;
7. refine by j_n , apply a selected unital spectator state, and verify Eq. (189) without fitting.
8. construct the source graph, verify finite propagation and the boundary-event response, and evaluate the all-stage Ward kernel in Eq. (167);
9. build the continuous certified face representatives, decompose the regular source by Eq. (169), and verify the exceptional positive quotient channels;
10. verify the continuous metric/nonmetricity field, active positive transport, and the ten Lorentz-metric response directions used in the v0.8 main theorem;
11. evaluate the finite action in Eq. (173), its coframe, connection, and metric variations, and the local-frame identity in Eq. (174);
12. verify the source-library balance no-go without replacing it by a fitted negative Euler residual;
13. evaluate the independent defect action in Eq. (176), its 832 exact variation checks, connector ranks, covariance tests, and Noether identities.
14. generate the one-site marker's exact spectral event projectors, replay the actual 25-event braid permutation and its refinement/readback controls, and distinguish this formal event law from an empirically calibrated physical event;
15. evaluate the ten finite response directions and the fixed full-corner matter packet against its declared partition controls;

16. replay the complete five-partition fixed-packet test, verify that only the full corner preserves the eleven-operator packet, and recover the normalized source cup 3/5 without target fitting;
17. reconstruct Q_4 , J_{ray} , and K_{evt} from the actual event transition, verify the exact $(3+, 1-)$ signature and unique S_4 -equivariant normalization, and replay the identity, flip, orientation, and closed-time-path controls.

This is already a nontrivial readout because it returns directional tensor data that scalar invariants provably cannot supply in the declared linear equivariant class. It is not yet an empirical observable because no instrument model, units, noise model, or identification with measured geometry has been given.

J.2 Formal falsifiers

The present theorem package would fail if any of the following occurred under exact replay:

- the target geometry entered the source constructor before comparison;
- either internal GL_3 presentation changed T_C ;
- A/B exchange changed the joint carrier;
- a one-sided collapse reproduced the joint carrier;
- the ambient carrier failed the congruence law;
- the all-stage readback required a tracial spectator state or a stage-dependent coefficient;
- the source-graph support propagated faster than the certified graph bound;
- the $3 + 1 + 5$ source product failed to reconstruct the registered geometric source;
- the source-first metric two-jet failed any algebraic Riemann symmetry, Bianchi, contraction, Lorentz-signature, or exact Einstein check under its declared bridges;
- the analytic action variations failed their exact central-difference checks;
- the local-frame Noether identity failed with all required Euler slots included;
- the covariant-defect action accessed the curvature residual or an Einstein target, failed to change off shell, or lost its stated connector rank;
- the exact marker spectrum, 25-event braid transition, five-child cylinder identity, or co-moving readback failed on a signed sector;
- any original F_A fell outside the declared event/energy readout span, or the full- W integration failed on a tilted source state;
- the RML identity or its historical 3/5 cup failed while the stated corner normalization and source ranks held;
- a non-full partition preserved the complete frozen event-conditioned packet, the selected normalized source cup failed to equal 3/5, or either result required comparison with the Einstein target;
- the actual-event Q_4 lost rank or covariance, K_{evt} lost its exact Lorentz signature, or the declared naturality class admitted a second positive normalizer;
- any registered exact regression ceased to reproduce its frozen source record.

J.3 Physical discriminators still required

To test the Creation Principle as a claim about nature, at least one bridge must produce an empirical discriminator. Four routes are open:

1. identify T_C with, or derive a calibrated map to, an operational metric or stress response;
2. test the frozen coefficient-one sum of the curvature and defect actions, then show that any resulting stationary equations possess a controlled continuum and low-energy sector;
3. derive a boundary/re-entry signal that standard quantum-reference-frame or effective-field descriptions do not predict;
4. show that a known physical structure is recovered with fewer independent primitive assumptions and with a novel constraint among measurable channels.

At the v0.8 boundary, the decisive remaining formal target was a finite Lorentzian quantum parent whose interaction stress supplied the missing spatial components of the discrete total Ward balance while preserving the already derived infrared sector. The v0.81 main text closes that target in the aligned four-score, block-local, history-dressed class and states the narrower remaining boundaries.

“Measuring what was previously unmeasurable” would count as a genuine contribution if the new readout has an independently specified instrument, calibration, error model, and discriminating result. A renamed internal variable would not.

K Objections and boundaries

K.1 Where is quantum theory?

The source is quantum in a specific algebraic sense: it uses complex noncommutative matrix algebras, states, unitary pointed-braid transports, tensor refinement, and adjoint observer actions. Those ingredients are enough to make the classical-output result nontrivial, but they are current inputs. The paper does not derive complex amplitudes, the Born rule, Hilbert-space composition, unitary time evolution, or quantum-field locality from a bare topological braid group. Accordingly, “a route to quantum structure” names the wider Braid-first programme. The exact result of this paper begins at a quantum-algebraic Braid source and derives the declared classical carrier and finite gravity dynamics from it.

K.2 Where is spacetime, and does the ambient four-space smuggle objectivity back in?

Version 0.5 already removed the numerical and ambient-carrier forms of this objection at the mathematical layer. The finite source produces $Q_{\text{obs}} = V_4$. Full affine-observer symmetry, an order unit, and point separation force the complete readout and hence the target-free split $1 \oplus 3$. Observer Uniform Cauchy Closure completes that carrier to $W = \mathbb{R}^{V_4}$ without adding a direction; $\text{Aff}(W)$ is then its own smooth four-dimensional affine carrier. The source path is not renamed as physical three-space, and the four metric-response controls are not renamed as region coordinates.

Version 0.6 added a checked finite event algebra and crossing/readback rule. The v0.6.1 code audit corrects the stronger conclusion drawn from it: BGCE138 constructs a smooth completion and compatible samples, but does not show that actual finite source fields converge to that completion. Post-v0.6.1 results retire the independent finite event-algebra correspondence inside the fixed model, but they do not prove the reverse source-to-smooth arrow. Thus formal CGR as a complete source-to-continuum statement remains open. Physical identification with measured events, dimensionful calibration, and continuum first variation for nature are additional requirements.

K.3 Is the carrier unique?

No global uniqueness theorem is claimed. Equations (179)–(181) are a natural, exact, source-native solution that obeys the declared symmetries and controls. Other nonlinear or higher-order carriers may exist. The counterexample to OP needs existence, not uniqueness. A stronger physical theory would benefit from deriving the carrier from an action, universality principle, minimality theorem, or experimental constraint.

K.4 Is a finite model enough?

It is enough to refute a universal mathematical necessity claim if the model is admitted. It is not enough to establish a fundamental theory of our universe. The nonterminal tower prevents the result from being only one isolated finite face, while the source graph, spectral events, and continuous certified cells add

mathematical notions of locality and curvature. Version 0.8 derives the infrared action, stress/Ward identity, and Einstein/spin-2 limit only in its declared universality class. Full finite Lorentzian quantum backreaction, ultraviolet completion, global extension, and empirical scale identification remain separate tasks.

K.5 Where is gravity?

The construction contains more than prerequisites for gravity. Equations (157) and (158) give an exact source-native coframe update and covariant curvature closure, while Eq. (159) gives the corresponding finite metric-affine update. Rank-changing geometry requires the source term, and every nonmetricity, source-linear, and source-quadratic contribution survives ablation. The same packet extends without refitting through all finite stages. The source graph supplies finite locality and a Ward law; continuous certified faces carry nonzero connection curvature; Eq. (173) supplies an exact first-order curvature action with Eq. (174) as its local-frame identity; and Eq. (176) supplies an independent off-shell variational connector. In this exact fixed-orbit sense, an action-bearing finite Braid metric-affine gravity candidate has been constructed.

The retained finite equations are the discrete geometric substrate. The v0.8 main theorem added a source-selected infrared action, continuous tensor transport, a unique typed solder, Hilbert stress, and on-shell Ward conservation, and then obtained $G_{\mu\nu} = (3/5)T_{\mu\nu}$ by common metric variation in the declared long-wavelength, local, second-order, formally self-adjoint, conservative class. Version 0.81 adds the scoped finite recorded parent, four-component history-dressed Ward law, and one finite backreaction cycle. Full ten-component nonlinear dynamics, gravitational constraints, physical units, ultraviolet completion, global extension, and empirical identification remain open. This is a Braid-based quantum-gravity candidate with exact finite results and a scoped Einstein derivation, not completed quantum gravity or source-only general relativity.

K.6 Is “subjectivity” only a relabeling?

It would be a relabeling if the formal roles made no difference. Here the roles are constrained: two prefixes must remain distinct; their internal presentation can be quotiented but their contributions cannot be collapsed; and their intersection produces a carrier with a verified transformation law. This is more than terminology. Nevertheless, calling these formal roles “subjectivities” is an interpretive move. The mathematics fixes the relational structure; ontology assigns its meaning.

Data, code, and provenance

The exact certificates and replay scripts are stored in the SIEL Research Agent repository, [SIEL-Research/SIEL-Research-Agent](#). Version 0.92 uses completion-ledger commit [50aca5a40](#). It retains the full v0.91 lineage and adds the post-v0.91 compiler, real-QPU, matter, and BQGEULER-004 records through BGCE569. The constraint closure is retained at [2096dfb4a](#), the anisotropic UCP closure at [98cb50608](#), and the all-finite-iteration backreaction closure at [934647260](#). The QPU packets preserve program manifests, provider task identities, raw counts, costs, controls, and negative statistical audits. The matter and convergence packets preserve exact witnesses, countermodels, validators, and claim ceilings. These are internal theoretical, computational, and exploratory operational certificates, not external replication, peer review, a Newton-constant prediction, general smooth-spacetime UV completion, execution of the full quantum actuator, or empirical quantum-gravity validation.

Declaration on computational assistance

OpenAI Codex was used under the author’s direction for mathematical ideation, exact-check scripting, audit organization, and drafting. The author retains responsibility for definitions, interpretation, verification, citations, and any decision to release or submit the work.

References

- [1] E. Artin, “Theory of braids,” *Annals of Mathematics* **48**(1), 101–126 (1947). doi:10.2307/1969218.
- [2] S.-i. Amari, *Information Geometry and Its Applications*. Springer (2016). doi:10.1007/978-4-431-55978-8.
- [3] C. A. Fuchs and R. Schack, “Quantum-Bayesian coherence,” *Reviews of Modern Physics* **85**, 1693–1715 (2013). doi:10.1103/RevModPhys.85.1693.
- [4] M. Fierz and W. Pauli, “On relativistic wave equations for particles of arbitrary spin in an electromagnetic field,” *Proceedings of the Royal Society A* **173**, 211–232 (1939). doi:10.1098/rspa.1939.0140.
- [5] F. Giacomini, E. Castro-Ruiz, and Č. Brukner, “Quantum mechanics and the covariance of physical laws in quantum reference frames,” *Nature Communications* **10**, 494 (2019). doi:10.1038/s41467-018-08155-0.
- [6] F. W. Hehl, J. D. McCrea, E. W. Mielke, and Y. Ne’eman, “Metric-affine gauge theory of gravity,” *Physics Reports* **258**, 1–171 (1995). doi:10.1016/0370-1573(94)00111-F.
- [7] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, “Completely positive dynamical semigroups of N -level systems,” *Journal of Mathematical Physics* **17**, 821–825 (1976). doi:10.1063/1.522979.
- [8] J. Ladyman, “What is structural realism?” *Studies in History and Philosophy of Science Part A* **29**(3), 409–424 (1998). doi:10.1016/S0039-3681(98)80129-5.
- [9] G. Lindblad, “On the generators of quantum dynamical semigroups,” *Communications in Mathematical Physics* **48**, 119–130 (1976). doi:10.1007/BF01608499.
- [10] M. Massimi, *Perspectival Realism*. Oxford University Press (2022). doi:10.1093/oso/9780197555620.001.0001.
- [11] C. Rovelli, “Relational quantum mechanics,” *International Journal of Theoretical Physics* **35**, 1637–1678 (1996). doi:10.1007/BF02302261.
- [12] W. F. Stinespring, “Positive functions on C^* -algebras,” *Proceedings of the American Mathematical Society* **6**, 211–216 (1955). doi:10.2307/2032342.
- [13] H. Umegaki, “Conditional expectation in an operator algebra, IV,” *Kodai Mathematical Seminar Reports* **14**, 59–85 (1962). doi:10.2996/kmj/1138844604.
- [14] A. Vanrietvelde, P. A. Höhn, F. Giacomini, and E. Castro-Ruiz, “A change of perspective: switching quantum reference frames via a perspective-neutral framework,” *Quantum* **4**, 225 (2020). doi:10.22331/q-2020-01-27-225.
- [15] H. Weyl, *Symmetry*. Princeton University Press (1952).
- [16] S. Watanabe, *Subjectivity-Intersection Mathematics: All-Orders Intersection Consistency and Compositional Objectivity from Pointed Unitary Braids*, version 1.0, Zenodo (2026). doi:10.5281/zenodo.22166167.
- [17] S. Watanabe, *Subjectivity Intersection Ontology: A Comparative Ontological Framework for Third Subjectivity*, version 1.0.0, Zenodo (2026). doi:10.5281/zenodo.21641878.
- [18] S. O. Bilson-Thompson, “A topological model of composite preons,” arXiv:hep-ph/0503213 (2005). doi:10.48550/arXiv.hep-ph/0503213.
- [19] P. Etingof, T. Schedler, and A. Soloviev, “Set-theoretical solutions to the quantum Yang–Baxter equation,” *Duke Mathematical Journal* **100**(2), 169–209 (1999). arXiv:math/9801047.

- [20] R. P. Feynman and F. L. Vernon, Jr., “The theory of a general quantum system interacting with a linear dissipative system,” *Annals of Physics* **24**, 118–173 (1963). doi:[10.1016/0003-4916\(63\)90068-X](https://doi.org/10.1016/0003-4916(63)90068-X).
- [21] E. Hawkins, F. Markopoulou, and H. Sahlmann, “Evolution in quantum causal histories,” *Classical and Quantum Gravity* **20**, 3839–3854 (2003). arXiv:[hep-th/0302111](https://arxiv.org/abs/hep-th/0302111).
- [22] P. Hayden, R. Jozsa, D. Petz, and A. Winter, “Structure of states which satisfy strong subadditivity of quantum entropy with equality,” *Communications in Mathematical Physics* **246**, 359–374 (2004). arXiv:[quant-ph/0304007](https://arxiv.org/abs/quant-ph/0304007).
- [23] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Physical Review Letters* **75**, 1260–1263 (1995). doi:[10.1103/PhysRevLett.75.1260](https://doi.org/10.1103/PhysRevLett.75.1260).
- [24] M. Massimi, “Epistemic communities and their situated practices: Perspectival realism—a primer,” *Annals of the New York Academy of Sciences* **1523**, 5–10 (2023). doi:[10.1111/nyas.14981](https://doi.org/10.1111/nyas.14981).
- [25] T. Nagel, *The View from Nowhere*. Oxford University Press (1986).
- [26] D. P. Rideout and R. D. Sorkin, “Classical sequential growth dynamics for causal sets,” *Physical Review D* **61**, 024002 (1999). doi:[10.1103/PhysRevD.61.024002](https://doi.org/10.1103/PhysRevD.61.024002).
- [27] R. M. Wald, *General Relativity*. University of Chicago Press (1984).
- [28] S. Watanabe, *Subjectivity-Intersection Mathematics Public Working Preprint v4.2 Draft*, Zenodo (2026). doi:[10.5281/zenodo.21909933](https://doi.org/10.5281/zenodo.21909933).
- [29] NIST, *2022 CODATA Recommended Values of the Fundamental Physical Constants*, 2022 release, accessed 25 September 2026. [NIST CODATA wall chart](https://www.nist.gov/codata).
- [30] BIPM, “SI defining constants,” accessed 25 September 2026. bipm.org/en/measurement-units.
- [31] Amazon Web Services, “Pulse control on Amazon Braket,” accessed 25 September 2026. [AWS Braket Developer Guide](https://aws.amazon.com/braket/developer-guide/).
- [32] Amazon Web Services, “Accessing native gates using pulses,” accessed 25 September 2026. [AWS Braket Developer Guide](https://aws.amazon.com/braket/developer-guide/).
- [33] A. M. Childs and N. Wiebe, “Hamiltonian simulation using linear combinations of unitary operations,” *Quantum Information & Computation* **12**, 901–924 (2012). arXiv:[1202.5822](https://arxiv.org/abs/1202.5822).
- [34] G. H. Low and I. L. Chuang, “Optimal Hamiltonian simulation by quantum signal processing,” *Physical Review Letters* **118**, 010501 (2017). doi:[10.1103/PhysRevLett.118.010501](https://doi.org/10.1103/PhysRevLett.118.010501).
- [35] A. Gilyén, Y. Su, G. H. Low, and N. Wiebe, “Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics,” in *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing*, 193–204 (2019). doi:[10.1145/3313276.3316366](https://doi.org/10.1145/3313276.3316366).
- [36] P. Høyer, “Arbitrary phases in quantum amplitude amplification,” *Physical Review A* **62**, 052304 (2000). doi:[10.1103/PhysRevA.62.052304](https://doi.org/10.1103/PhysRevA.62.052304).
- [37] G. Brassard, P. Høyer, M. Mosca, and A. Tapp, “Quantum amplitude amplification and estimation,” in *Contemporary Mathematics* **305**, 53–74 (2002). doi:[10.1090/conm/305/05215](https://doi.org/10.1090/conm/305/05215).

Question	v0.92 result	Remaining boundary
Operational four-direction carrier	Retained exactly from the source cylinder and four null links	No universal four-dimensional uniqueness theorem
Matter realization and 3/5	Fixed-model MMR2 closed; normalized source-cup ratio derived	Arbitrary instruments and SI-unit calibration remain open
Lorentz quantum parent	Local raw/Petz-recorded CTP parent on the source cylinder	Aligned four-score and block-local scope; full ten-component nonlinear parent open
Continuous tensor field	S4 gluing and passive local $GL(4)$ naturality exact	Empirical identification with natural spacetime remains open
Physical stress and Ward law	Branch contrast derives all four transfer defects; exact history-dressed finite Ward	Arbitrary smooth diffeomorphism Ward and spatial-gradient dynamics open
Einstein and spin-2 limit	Noncircular scoped derivation with two helicities	Finite gain 1 and routing weight 3/5 are differently typed; physical calibration remains scoped
Finite backreaction	Every finite operator-logistic identity-feedback iteration remains CPTP, Ward-compatible, rank four, and nonlinear	Dynamics not unique; no $n \rightarrow \infty$ uniform bound or continuum strong-curvature theorem
Matter content	Three chiral SM character copies, one Higgs doublet, a nonzero finite Higgs minimum, variational relative Yukawa, and finite spectra	Observed labels, universal scale, RG flow, physical mixing, Majorana scale, and empirical confirmation open
Constraint algebra	Braid-specific rank-three closable unbounded local derivations; generated finite-order algebra closes with exact Jacobi identity	No ADM moment-map typing or hypersurface-deformation identification
UV and all-scale control	AF/quasi-local limit, rank-35 UB612 response, and actual-metric anisotropic hierarchical UCP heat/positive wave with spectral dimension three	Conductance uniqueness, smooth principal symbol, composite-stress renormalization, and general physical UV completion open
Physical calibration	E2-audited exploratory AWS frequency anchor and CODATA-input conditional stress scale	Provider-clock SI traceability and complete uncertainty budget absent; G is input
Quantum kinematics and probability	A 36-dimensional complex carrier, source-dagger Born probabilities, exact $\mathbb{Z}_2$ superselection, and $M_{18}(\mathbb{C}) \oplus M_{18}(\mathbb{C})$ observables	Finite source-dagger scope; ordinary flip shares the complex-sector mechanism; no empirical identification
Finite quantum control	Exact Chebyshev actuator and a 34-CNOT phase-frame source generator executed on real QPU hardware	The full 203–39,102-query actuator was not executed; no quantum advantage or gravity measurement
Continuum wave dynamics	Two-polarization hierarchical wave trajectories converge strongly in energy norm uniformly for all real time	Physical ADM projector, smooth Fierz–Pauli symbol, and nonlinear Euler convergence open
Higgs and fermion sector	Nonzero finite Higgs minimum, variational relative Yukawa operator, and source/device-relative bare and dressed spectra	Observed labels, universal scale, RG flow, total uncertainty, and empirical matching open
Neutrino extension	Three neutral carriers and $y_\nu = +1$ inside a declared unital Frobenius-positive completion	Unchanged-source particle typing and dimensionful Majorana scale not derived

Table 1: The v0.92 dependency split. “Derived” always means derived inside the declared fixed source model or explicitly derived extension. “Calibrated” is limited to the device-relative and conditional external-input scope stated here.